NANA SHILPA: SCREENING AND REFINEMENT SYSTEM TO IDENTIFY THE RISK OF DYSCALCULIA AND DYSGRAPHIA LEARNING DISABILITIES IN PRIMARY SCHOOL STUDENTS

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Dissertation submitted in partial fulfilment of the requirements for the Bachelor of Science Special Honors Degree in Information Technology

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DECLARATION

We declare that this our own work & this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning & to the best of our knowledge & belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Learning disability is a condition that has a direct effect on the brain and there is no cure or any identified medical treatments. Students with learning disabilities manifest weaknesses in some specified areas, compared to their fellow peers. The majority of these cases remain undiagnosed due to the lack of awareness from their parents and teachers. Even with the rapid improvement of technology, limited attention has been given to the diagnosed students in Sri Lanka. Through this research, a mobile-based solution named 'Nanashipla' was developed for the screening and intervention processes for the specific Learning Disabilities which are Verbal and Lexical Dyscalculia, Operational and Practognostic Dyscalculia, Letter Level Dysgraphia and Numeric Dysgraphia. Both screening and intervention processes are provided by the mobile application. In order to provide a better solution, Machine Learning and Deep Learning methods are used in the screening process. Convolutional Neural Networks (CNN) were utilized to detect the written letters and numbers. It is achieved the accuracy of 92%, 99%, 99% for Verbal and Lexical Dyscalculia, Letter Level Dysgraphia and Number Dysgraphia Convolutional Neural Networks respectively. Machine Learning models are trained using the outputs from Convolutional Neural Networks. The Machine Learning algorithms used in the Machine Learning models of screening processes are Support Vector Machine (SVM) and Random Forest (RF). In the machine learning models, it is achieved the accuracy of 98%, 97% for Operational and Practognostic Dyscalculia and Number Dysgraphia respectively. In Sri Lanka, this has been recognized as an acceptable solution for screening and intervention via a mobile based application for Verbal and Lexical Dyscalculia, Operational and Practognostic Dyscalculia, Letter Level Dysgraphia and Numeric Dysgraphia which is developed based on the gaming environment.

Keywords—Dyscalculia, Dysgraphia, Machine Learning, Convolutional Neural Network, Mobile application, Learning Disabilities

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LIST OF ABBREVIATIONS

Abbreviation	Description
API	Application Program Interface
ADHD	Attention-Deficit/Hyperactivity Disorder
LD	Learning Disability
ML	Machine Learning
SVM	Support Vector Machine
ROC	Receiver operating characteristic curve
MOH	Ministry of Health
ND	Numeric Dysgraphia
RF	Random Forest
CNN	Convolutional Neural Network
NT	Not Tested
RBF	Radial Basis Function
MNIST	Modified National Institute of Standards and Technology database
ReLu	Rectified Linear Unit
1D	One Dimension
MFCC	Mel Frequency Cepstrum Coefficient
JSON	Java Script Object Notation
3D	Three Dimension

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1 INTRODUCTION

1.1 Background

Learning disabilities (LDs) itself is not a very cordial term for under developing countries like Sri Lanka. Due to the lack of awareness and knowledge about learning disabilities, many students face discomfort in their learning process. In the majority of situations, students with learning disabilities are eliminated from further educational opportunities and neglected from achieving the learning outcomes. Even though Sri Lanka has achieved the 99.11% in the primary school enrollment rate as reported by [1], we have yet to acquire a long journey in establishing a quality inclusive education where students with disabilities can be identified and treated in the early stages refer to [2]. The accurate rates of students who are undergoing learning disabilities cannot be found since many cases of the disability occurrences are not verified or identified medically.

Due to these learning disabilities, students may develop aggressive behaviour, school violence acts or victims of school bullying also. Refer to [3] learning disabilities had increased the risk of getting bullied by their peers and also social, emotional and behavioural damage which affects their growth and development of life.

These challenges can be addressed through a conducive learning environment. Through this research we have approached, an assistive Learning Environment that will improve the learning experience of LD students in their academic life. Learning handicaps stem from defects in one or more understanding, reasoning, recalling or learning-related processes which include language processing; phonological processing; space visual processing; the speed of processing; memory and attention and executive function. Concerning [4], the education metrics, Sri Lanka almost achieved the Millennium Development Goals (MDGs). Given 98 percent of elementary school enrolment (World Bank, 2013), however, many disabled children

tend to be excluded and still remain as the subjects of high attrition levels. For example, 23.5% of school children (5-14 years of age) with handicaps are excluded from education [4].

Dysgraphia is a neurological disability condition where the students show unclear and irregular handwriting, difficulty in writing in provided space and slow writing compared to their peers. Refer to [4] the cause of this disability yet to discover but it has been stated that it is the result of the damage of the parietal lobe of the brain.

According to [5],[6],[8],[9], Dyscalculia is a term that can be described as a specific Learning Disability affecting the ability of an individual to understand numbers and discover more in mathematics. Dyscalculia is sometimes called number blindness. Much of the work into the development of mathematical skills has centered upon counting or arithmetic, but other complex skills also require these skills at early levels including: [5] knowing the word number (one, two, ...), numerals (1, …12, … 20, …) and their interrelationships. As stated by [7], Dyscalculia has distinction inside itself. Lexical dyscalculia is the variant that causes trouble in reading mathematical symbols. Verbal Dyscalculia is the variants that trouble in name mathematical quantities, numbers, and symbols. Operational Dyscalculia condition causes troubles in performing mathematical operations and calculus and Practognostic Dyscalculia causes troubles in enumerating, manipulating, and comparing real objects and pictures. These 4 variants of Dyscalculia have been used in the application in the screening and intervention process of Dyscalculia.

This research project was conducted to develop a mobile application for screening and intervention in the risk of dysgraphia and dyscalculia learning disability in primary school students in Sri Lanka. The fundamental objective of delivering this mobile application is to cater the accessible screening and intervention method for these risk

of learning disabilities and elevate the basic skills of mathematical operations in calculations, reading, and Sinhala and number writing in primary school students.

1.2 Literature Survey

Some researchers have been conducted in recent years to develop an assistive tool for identifying dyscalculia in children. When going through the past researches which were conducted about Dyscalculia, the "Dyscalculia Screener"[10],[11] is a program designed to evaluate the mathematical capacity of children 6-14 years old. It checks the children's wicker zones by giving various questions such as arithmetic operations in mathematics, comparison. But "Dyscalculia Screener" does not support both screenings and provides a solution to overcome this learning disability. But the limitation was it is only supporting screening. In [8], the proposed solution to considered problems was a client-server web application which was used HTML5, CSS, JavaScript while using MySQL as the database during the implementation. During the research, students had been checked calculations, numeric concepts, and problem-solving areas.

"DyscalculiUm" [12] is another screening test based on a computer that evaluates children with Dyscalculia. It was designed as a web-based solution to detect the risk of Dyscalculia but, it assesses students who take part in higher education only. For the screening section for this diagnose, it was used six key factors to identify. They are numerical and mathematical concepts, fractions, decimal, negative numbers, money, and time. Test items were made to prove that. Refer to [13], they have developed five different cognitive tasks to screen Dyscalculia patterns associated with two systems, and screening test is done by using a tablet to mobilize the tool to enable the data collection.

Among many numbers of other tools, the “Dynamo Maths” [14] is computer software that can be used for intervention and remedying people with Dyscalculia. It is

developed for children who are between 6 and 9 years old. The three-stage program includes lesson plans on multi-sensory resources, interactive online modules, and worksheets for practice. According to [15], research which was done in Thailand, the main purpose was to develop assistive technology for the students who suffer from learning disabilities mainly Dyscalculia. At the end of this research, it had been provided with a software tool named “Calculating Aid Tools: KidKanit” helps Dyscalculia students in mathematical areas.

The study [17], proposed a web-based learning tool as a design to improve the learning environment of different technologies of Dyscalculia students. In the beginning, the research has identified a fact that there is a chance to 50-60 percentage of dyslexia people has suffered from Dyscalculia also. it was only provided simple and very attractive game-based activities for the pre-school students. But there was no dyscalculia identification process. The free online screening tool called "Dyscalc" is designed for students over 14 years of age [16]. Further, it is a web-based solution.

Research in [13], a quick (2 minute) paper-and-pencil tool was developed to evaluate the ability of children to compare symbolic and non-symbolic numerical magnitudes to resolve these outstanding problems and evaluated the degree to which individual differences in achievement are explained by success on this test.

In [20], a study was done for the diagnosis of dyscalculia using three classifiers rule-PART, J48, and SVM to predict the different classes of Dyscalculia. The final result was got based on attributes such as difficulty with shape recognition, difficulty with size discrimination, number arrangement, numeric calculations, counting concepts scores. Three models were created using collected data and founded SVM had better resulted in the classification of various Dyscalculia subclasses with a higher accuracy value is 94% compared to the other two classifiers.

Most of the existing researches done and developed applications as a web-based solution [8],[17],[18]. But nowadays the most effective way of learning things using mobile [19]. Refer to, [5] mobile learning is an important way for Learning Disability children to do math exercises and enhance the memory to memorize the math concepts. It is therefore the most successful way to develop an application as a mobile app.

Another research [21] called “Calculic Kids” was designed to help Dyscalculic children in the numeric learning-based mobile application learning method. This application developed using the Android platform with support for their native language. According to them, a mobile application will help the student to learn math effectively and grab the concepts quickly. They gathered requirements through interviews with the program coordinator and math teacher. Four subsets of games developed for number, object counting, addition, subtraction. The drawback of this application is, no remedy activities for users and only a series of games provided. Also, there is no way of tracking the user's progress.

According to [22], research work of developing an intelligent system to diagnose and suggest remedial measures for learning disorders in children leads to identify common learning disorders in children. Target persons for the developed system are parents, teachers, researchers. Based on the input provided by the parents about their child’s behaviour, system diagnoses the disorders and suggest the remedial measure. The technology used by them to develop the system is C#. The system consists of 3 modules called knowledge base, inference engine, and user interfaces. Configuration module garners all pertinent information from other users in an interactive mode. All queries are in Yes or No format. The main aspect of this system is, it collects child behaviour through a questionary by their parents to diagnose disorders and provide remedies.

The role of the mobile application developed in research [23] is to improve the hard life of persons with learning disabilities. The application supports individuals with learning disabilities to practice arithmetic and language disabilities to improve their brain efficiency. Language session, Arithmetic session, diagnosis, and test modules are provided to their users. Addition and subtraction, multiplication and division, reasoning, mathematics are the main types of lessons in the application. Enhance the application with any age by using EEG analysis algorithms is their suggestion for future work.

As mentioned in [24], the aim of examining the effects of computer-assisted instruction materials on the approximate number of skills in students with mathematics learning difficulties addressed by this research. The contents of the computer-aided application designed consist of counting skills, place value concepts, and additional subjects which are related to mathematics lesson's learning outcomes of primary school children. Dyscalculia screening tool and Panamath program were used as data collection tools by researchers.

In [25], mentioned research was conducted about the Dysgraphia situation of the students and had been provided mobile and web-based software tool that contains various tasks of writing practices. The tool was developed based on some difficulty levels for the students and it can be used on any mobile devices such as tablets, smart phones. PHP with HTML, CSS or JavaScript are the technologies used to develop this tool while MySQL was chosen as the database. In the implementation, Machine learning algorithms have been used such as lightweight recognizer and Time Warping algorithm. In evaluating phase, this application provides a report. Only the screening process was provided from this research.

Based on the scientific study [26], it was conducted about the students who show symptoms of Dysgraphia and not able to adjust to the learning process as their fellow

peers. The main objective was to implement an expert system, to help the diagnosed students with their studies and other activities. This expert system was computer software and it was implemented using Forward Chaining Method (FCM) which used for moving forward for getting the solution by considering the facts as a special characteristic. A written task is provided as the screening process which is given at a certain time. A timer was used to catch the time taken as a stopwatch. If a student can finish the task in the given time, some questions will be asked. End of the screening process, results will be divided into 3 components which are Normal, Light Dysgraphia, Moderate or Severe Dysgraphia. Only the screening process was provided from this research.

According to [27], this research was about to identify Dysgraphia students who need to improve many motor skills and processing skills. As the solution, AR-based Dysgraphia Assistance Writing Environment (AR-DAWE) modal was developed which addressed spelling mistake issues. This model provided real-time spelling assistance based on the voice input. After that, real-time spelling feedback will be provided to the students. Simply, through this research AR-DAWE model converts the input which was user voice to texts using the cloud service and then shows the correct spellings through using the AR generated content. This model helps only the spelling issues of the students.

In [28], this research is conducted to identify handwriting issues of the Dysgraphia students. The main objective of this component was to provide an automated and complex solution for quantification for different Dysgraphia domains. A filter-based algorithm, minimum Redundancy Maximum Relevance (mRMR) algorithm, Sequential floating Forward Selection (SFFS) algorithm, ensemble learning algorithm such as RF (Random Forest) were used during the implementations. A practical questionnaire is used for the screening process which is the only functionality provided from this research.

Based on [29], this research is conducted to identify the dysgraphia students based on characteristics of their performance when compared to their peers. For the screening process, a questionnaire called standardized Handwriting Proficiency Screening Questionnaire (HPSQ) and as the environment Computerized Penmanship Evaluation Tool (Compet) as a digitizer tab and digital pen. The screening process contains four main tasks including write a 6-word sentence and repeat a letter 7 to 10 times. The result was analyzed based on the writing pattern and getting the pressure on the pen.

According to [30], a game was implemented using the client-server web application architecture. In order to maintain the interest level of students, the game-based approach was used and the game content of this system is that “runner player” game where the user collects the different types of fruits that appear on the screen which carries their respective game points. On the journey of the user, there are obstacles which need to be eliminated by drawing different shapes like circles, line. The user interfaces of the system were designed in a way, where the student who has not the ability to read should also be able to play the game. The key design principles which mainly focused were improved security for the user, interactive and informative guidance, high availability.

In [31], innovative approach to the system where user can view the performance of each student user of the system or overall view of the entire class of the students which are allocated to them. There are two separate interfaces for the student user and teacher, in order to maintain the integrity of the data and performance.

1.3 Background Survey

In order to get a clear idea about the awareness and importance of developing a system for identifying the risk of Dyscalculia and Dysgraphia learning disability in primary school students, an online survey was conducted. The survey was conducted by using two types of participants groups which were primary school teachers (Figure 1.1) of Homagama Educational Zone and the general public which were covered multiple age groups (Figure 1.2). The extracted results of the survey are listed below.

From these survey results, we have come to a few conclusions which aided to obtain the below information.

- Many of the teachers and the general public in society doesn't have a sound idea or awareness about learning disabilities.
- But in most scenarios, they have either experienced or witnessed the situations which were possible outcomes of having a learning disability.
- Also, they are seeming to be interested in the idea of developing a mobile application-based system to identify the risk of these learning disability conditions.

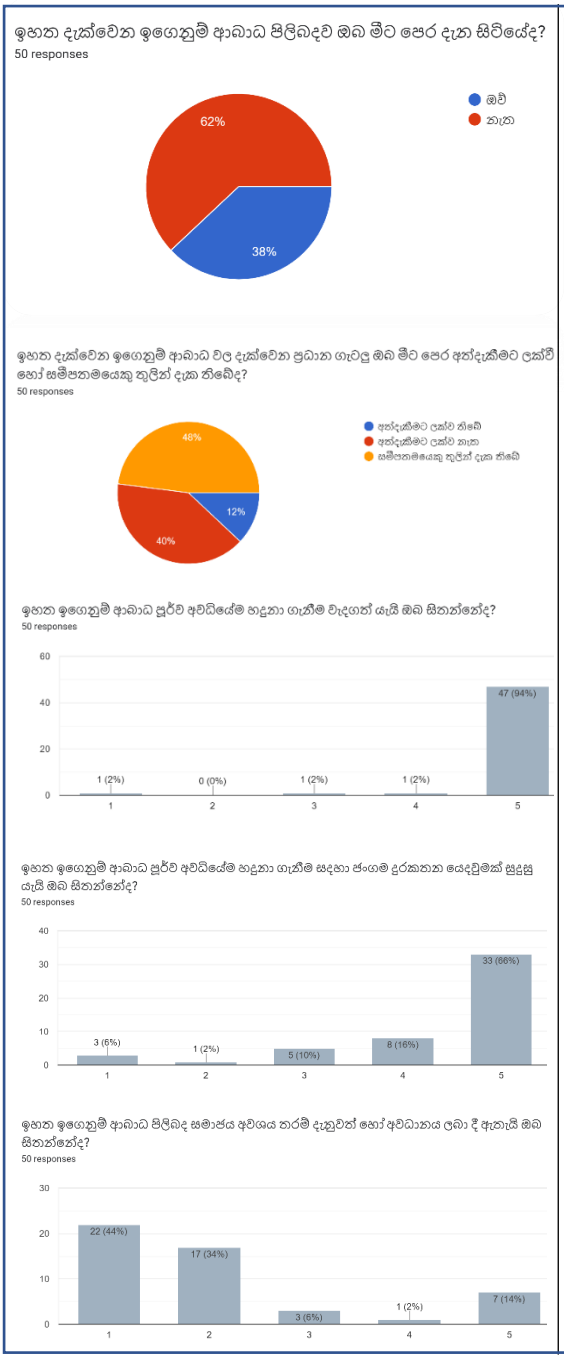


Figure 1.2: Results from Teachers

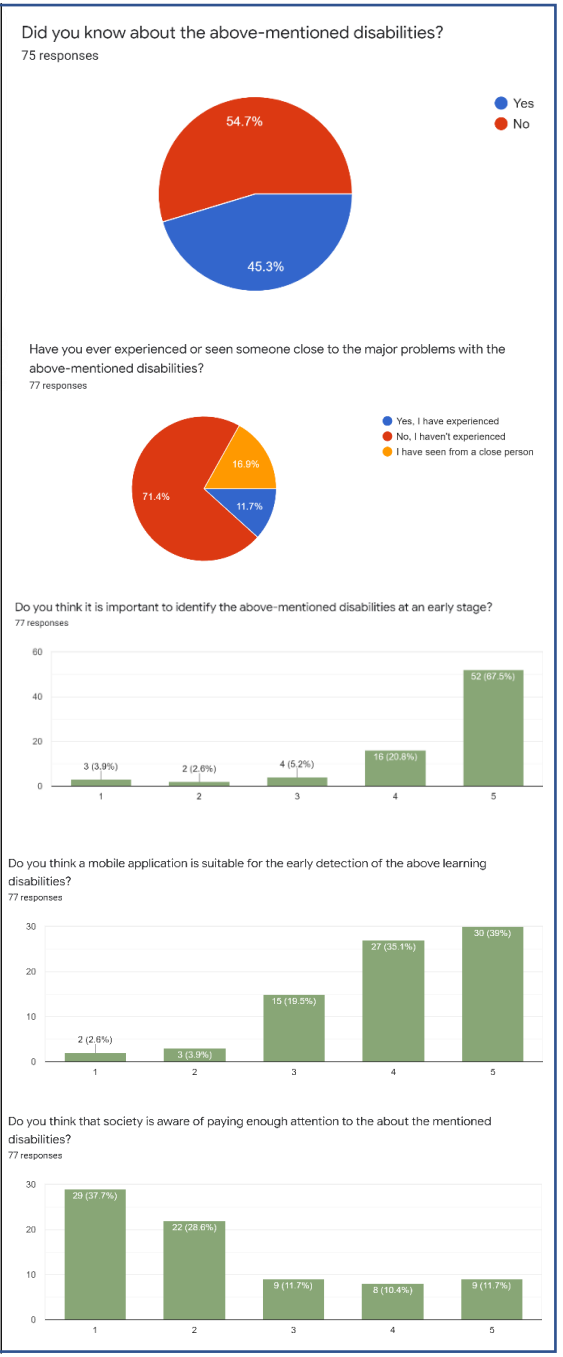


Figure 1.1: Results from General Public

1.4 Research Gap

Many present applications which were related to the domain had not supported both screening and intervention for mentioned two disability conditions. And significantly none of them had implemented to support local Sinhala langue and they had failed to deliver accurate results through their applications. Due to the major absence of these features in the Sri Lankan context, these applications cannot generate a positive impact on the students.

Table 1.1: Comparison of existing systems

Features	[8]	[15]	[17]	[26]	[29]	"Nana Shilpa"
Dyscalculic Identification.	✗	✗	✗	✗	✗	✓
Dysgraphia Identification.	✗	✗	✗	✓	✓	✓
Provide activities to improve the math concepts.	✓	✓	✓	✗	✗	✓
Online / offline access	✗	✗	✗	✗	✗	✓
Provide practice activities to improve the writing skills.	✗	✗	✗	✗	✗	✓
Support for Sinhala Language	✗	✗	✗	✗	✗	✓
User attractive interfaces with sound bases (multi-scenery)	✗	✓	✗	✗	✗	✓

Keep track of progress						
Report Generation						
Uses machine learning and Deep learning technique.						
Voice recognition techniques						

1.5 Research Problem

In under developing countries like Sri Lanka learning disability condition in primary school students are not identified or treated properly on a great no of occasions due to the lack of awareness among the teachers and parents. In the literature survey we had conducted, it has been observed out that there were not many applications that had been implemented to identify the risk of Dyscalculia and Dysgraphia Learning Disabilities in Primary School Students using the novel machine learning and deep learning techniques. Also, none of them had addressed the process of providing the intervention for the students through the application.

- Medical examination sessions for learning disability conditions are expensive and time-consuming.

Normal medical examination session with the child physiatrist who is specialized in mental disability conditions can cost more than 6000 Sri Lankan rupees. For the majority of Sri Lankan low-income families, this much expense can't bear due to their day to day living expenses.

- Absence of better and interactive user-friendly environment in the application.

Since the target age group is 6 to 8, it is very important to ensure that child is interested, and the child is taking the screening test without feeling awkward. For that, the application must provide an interactive and multisensory environment. But the existing applications had failed to produce that kind of environment in their applications.

- Deficiency of novel technology concepts in applications.

The majority of the existing applications are not practiced and implemented using novel and featured technologies like machine learning and deep learning in order to generate more accurate and reliable results.

- Not supported local language.

The significant issue of the existing applications is none of them has supported the native Sri Lankan Sinhala langue in their applications. Since the Sinhala language is the primary educational language in Sri Lankan educational system, it is vital to support the screening process, interventions, and inbuilt process navigations in Sinhala.

After the consideration of the above-mentioned complications, the main identified research problems which are required to address through this research project are how to provide a mobile application-based solution to screen and predict the risk of Dyscalculia and Dysgraphia learning disabilities in primary school students and deliver Comprehensive activities to overcome those learning disability risks.

2 OBJECTIVES

2.1 Main Objective

The main objective of this research is to provide a solution which is a screening and refinement system for primary school students aged 6 – 8. The solution is a mobile application that contains screening and intervention for Verbal Dyscalculia, Operational and Practognostic Dyscalculia, Letter level Dysgraphia and Numeric Dysgraphia. Since all these mentioned Learning Disabilities are uncurable through medicine, only active practices are needed to improve the condition. For that, identifying at the early stage is very important. Based on this concept, the solution is provided using the latest techniques such as deep learning and machine learning algorithms. The final product is provided in a user-friendly, and less expensive environment to build up the basic skills such as writing, reading, understanding, and performing skills of students.

2.1 Specific Objectives

From the main objective, there are four specific objectives were derived in order to provide a completed solution at the end of this research. All the mentioned screening and intervention activities below are followed under direct guidance from medical professionals.

- Identify the risk for Operational and Practognostic Dyscalculia and provide activities to improve the condition.

The implemented 'Nana Shilpa' application provided both screening and intervention options for Operational and Practognostic Dyscalculia. In order to identify the risk, the implemented application contained a mathematical problem-solving activity series based on the basic mathematical concepts including enumerating, operators and comparison. As the student faces the screening test series, the test results are shown

with the Dyscalculia situation as a probability. If the student turned out to be having a Dyscalculia situation, the intervention also provided. As the intervention, a set of comprehensive activities are provided by the application using a multisensory environment.

- Identify the risk for Lexical and Verbal Dyscalculia and provide activities to improve the condition.

For the Lexical and Verbal Dyscalculia also, the 'Nana Shilpa' mobile application is supported both screening and refinement activities. For the screening option of Verbal Dyscalculia, a selection activity is performed. Here, the student has to select the relevant numbers based on the prior voice cuts provided by the application. In the end, Verbal Dyscalculia prediction is done based on the correctness of done questions in the activity. To identify the Lexical Dyscalculia, a pronunciation activity series is conducted as the screening option. When the student reads the number shown in the application, it is recorded as a voice clip. So, in the end, the final prediction for Lexical Dyscalculia is shown based on considering the accuracy of the accent and the pronunciation based on the recorded voice clips of reading numbers by the student. If in both categories, the prediction is shown as there is a risk, the intervention also developed. Here, after the application shows a voice-guided number, the student should say it aloud, and then select the relevant number on a selection activity.

- Identify the risk for Letter level Dysgraphia and provide activities to improve the condition.

Implemented 'Nana Shilpa' application contains the ability to screen the risk for Letter level Dysgraphia and then providing intervention also. To screen the Letter level Dysgraphia, a writing test activity including letters and words is conducted through the application. The result is predicted as a risk for diagnosing Letter level Dysgraphia based on the accuracy of written letters and words. If the student is notified that there is a risk, the intervention activity also developed. It is also a test activity for writing

letters with words with prior guidance from the application. Only the student has to follow what is stated in the guidance.

- Identify the risk for Numeric Dysgraphia and provide activities to improve the condition.

Numeric Dysgraphia is a situation where students cannot write numbers accurately. So, through the developed 'Nana Shilpa' application, the screening for Numeric Dysgraphia as a risk and then provide remedies were implemented. First, identification for Numeric Dysgraphia is needed. So, a number writing screening test is performed. At the end of the screening quiz, the result is shown as a risk for having Numeric Dysgraphia based on the factors such as total correctness and total erase count which was taken while doing the quiz. If the student is warned as having Numeric Dysgraphia by the application, the intervention also included. It is provided as guidance-based activity. Here, also student can practice number writing with prior guidance.

- Keep the track of progress.

All the done screening quiz results of Operational and Practognostic Dyscalculia, Lexical and Verbal Dyscalculia, Letter level Dysgraphia and Numeric Dysgraphia are stored and shown by the implemented 'Nana Shilpa' application separately and clearly. So, a parent or a teacher can have an idea about the progress of a diagnosed students by viewing done quiz results.

3 METHODOLOGY

3.1 System Overview

The main objective of this application is to provide a mobile application-based solution to screen and predict the risk of Dyscalculia and Dysgraphia learning disabilities in primary school students. The "Nana Shilpa" mobile application system conducts the screening and refinement process of risk of Dyscalculia and Dysgraphia disability condition using 4 major risk identification and intervention components which are Verbal Dyscalculia, Operational and Practognostic Dyscalculia, Letter level Dysgraphia and Numeric Dysgraphia. Figure 3.1 illustrates, the system overview diagram of the Nana Shilpa system.

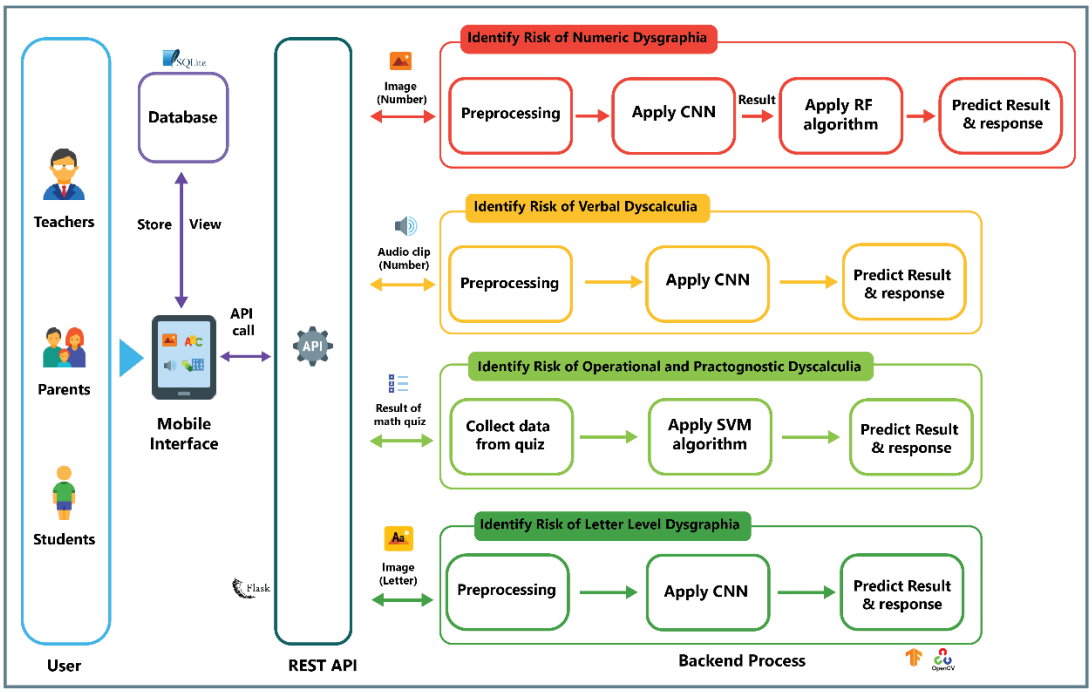


Figure 3.1: System overview diagram

3.2 System Methodology

3.2.1 Identify Risk of Letter Level Dysgraphia

The Letter Level Dysgraphia screening process has been conducted using four letters and four words in two stages. These four words and four letters were selected on consultation with primary school teachers and teachers guide of Sri Lanka local Sinhala syllabus. The selection of letters and words were ensured that it enfolds all the sections of the alphabet. In order to make the screening writing process more accurate and accomplished, students are asked to write letters and words at a time, and they are facilitated with clear the canvas and undo the movements features at the screening activity user interface as shown in Figure 3.2.



(a) Letter screening activity (b) Word screening activity

Figure 3.2: Letter Level Dysgraphia screening

One of the major challenges of the implementation was the absence of the handwritten Sinhala Letter and Word dataset. To overcome this challenge, more than 1000 handwritten letters and words were collected from school students which were located in multiple regions across Sri Lanka. The letter and word data collection had been processed by participating the students who are in the age range of six to eight using the built-in canvas in the Viber application.

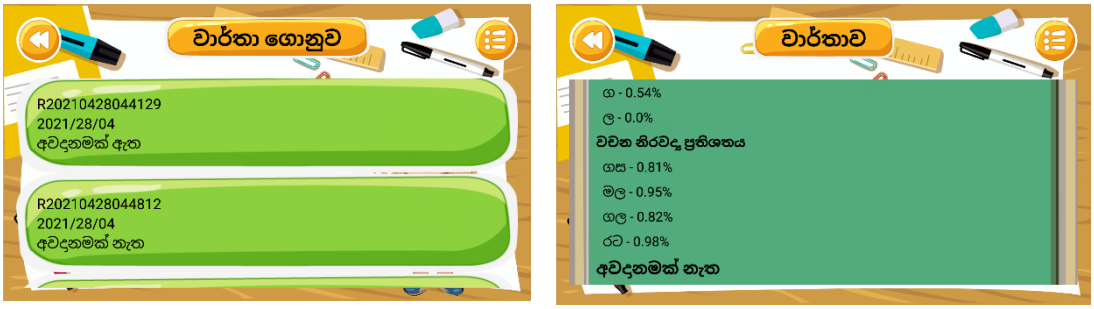
Initially, the handwritten letter images (.png file type) undergoing with image binarization process which results in the binary images. The purpose of this process is to obtain the binary image which contains two pixels values that are relevant to black and white. Since the image collection process had been carried out in different variant screen devices, the binary images were yelled to the image normalization process to obtain a constant image size of 32*32 using the resize function of the numpy library.

The Convolutional Neural Network model (CNN) was built using the letter and word data images which were accumulated from both dysgraphic and non-dysgraphic students who are in the age group of 6 to 8. The model consists of 4 conv2D layers, 2 MaxPooling2D layers, a flatten layer and 2 dense layers. The conv2D layers contain 64 filters with a kernel size of 5x5 and 3x3 strides. The maxpooling2D layer contains a pool that has the size of two along with two strides. Also, it combines a dense layer including 500 units. The Adam optimizer was determined as the optimizer function for the model considering it generates great results for the optimization of the gradient descent and solving the complex problem which involves a large no of data and parameters. Categorical cross-entropy loss used as the binary classification loss function as it depreciates the distance in between the actual and predicted probability distribution. With an additional utilization process, a trained CNN model is used for the accurate screening of the risk of letter level dysgraphia condition.

In the screening process that involves the letter and word identification through the CNN model, initially, the image is being preprocessed by converting it to greyscale, contrast is improved by applying Histogram Equalization and undergo with the binarization process. Meanwhile, in the segmentation process, letters and words are set apart. During the screening process, a REST API request has been sent to the backend server using the flask API which contains the user handwritten letter and the index value allocated to that distinct letter. Then using the trained CNN model built-in functions which are predict and predict classes, probability value regard to the image and class index value are generated to that relevant image. If the index value and

generated class index value is not equal the probability value is set to the negative value in order to indicate that the handwritten letter is not accurate for the expected letter input.

For every screening test attempt, related results which include the screening test date, accuracy percentage of each letter and words and final result whether the user has the possible risk level are being stored in the SQLite database as the report for further reference of the parent s and teachers. As shown in Figure 3.3 user can select the report record from the report archive and view the detailed results of that selected screening test. The most critical section of this component is the screening process of the handwritten letters and words and it adds significant value for the Nana Shilpa mobile application.



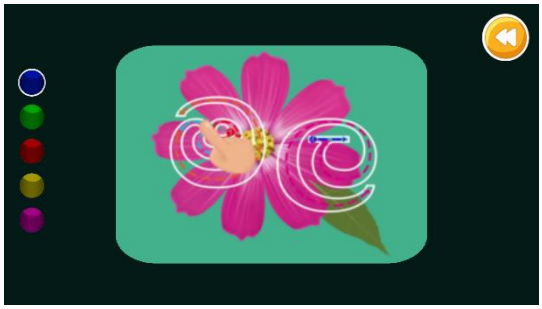
(a) Screening test report archive (b) Detailed report view of the screening test

Figure 3.3: Letter Level Dysgraphia Reports

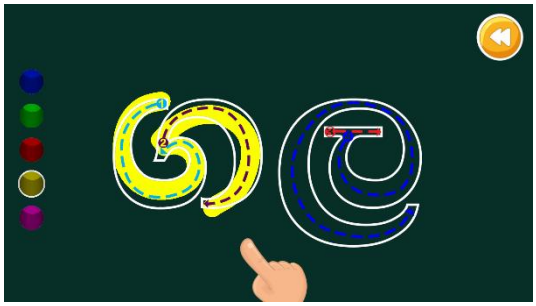
Upon the result of having the risk of letter level dysgraphia, extensive intervention activities are provided to the student. The intervention activities are provided in two stages which are for intervention for letters and words as shown in Figure 3.4.a. Due to the complexity of the Sinhala letters and words intervention practice activity provides guidance to the user in the multiple stages in the process.



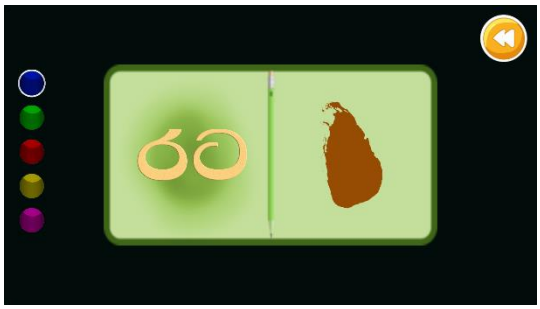
(a) Sub menu interface for letter intervention



(b) Demo video guidance for intervention actitvity



(c) Intervention practice UI



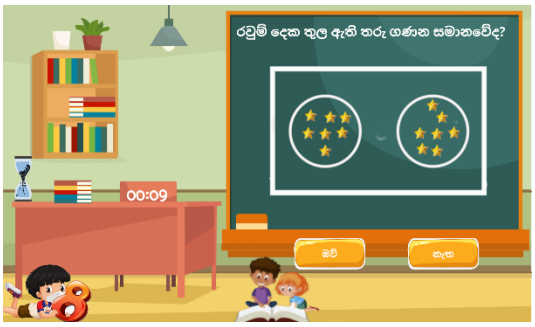
(d) Success popup for intervention activity

Figure 3.4: Letter Level Dysgraphia Intervention

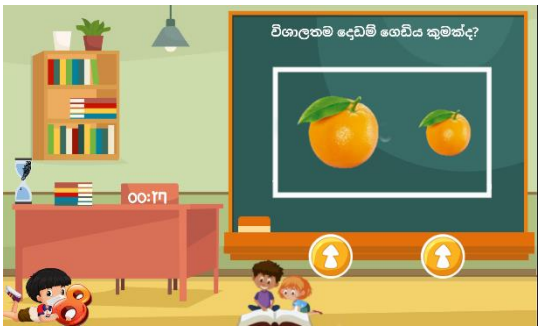
As seen in Fig 3.4.b when the user proceeds with the letter which they want to practice, initially a demonstration video clip that shows how to write the letter accurately. In the user interface of the intervention practice activity, the letter and words are guided with the dashed line and order of the steps in multiple color combination which helps the user to identify the user the proper order of writing. In each step, the letter is filled only if the user writes the letter accurately in the dashed line path shown in Fig. 3.4.c. As in Fig. 3.4.d, only after the user writes all the steps correctly which are allocated to the letter, a popup message appears with the chosen letter and the image related to the letter. The intervention process has implemented using Unity due to its flexible cross-platform deployment support and ability to develop a variety of graphics. The total process is conducted with background voice guidance and a mission-based story. The process is designed to obtain a more user-friendly and interoceptive learning process in which students can experience more positive outcomes.

3.2.2 Identify Risk of Operational & Practognostic Dyscalculia

In Operational and Practognostic Dyscalculia screening, an interactive mathematical quiz covering various areas was carried out to identify the children who suffer from Dyscalculia or not. Initially, as shown in Figure 3.5.a enumerating (counting) quiz were given. As a second and third stage, users have to complete the comparison of numbers and objects quiz as well as the mathematical operation-based quiz covering the addition and subtraction respectively as shown in Figure 3.5.b, Figure 3.5.c, and Figure 3.5.d.



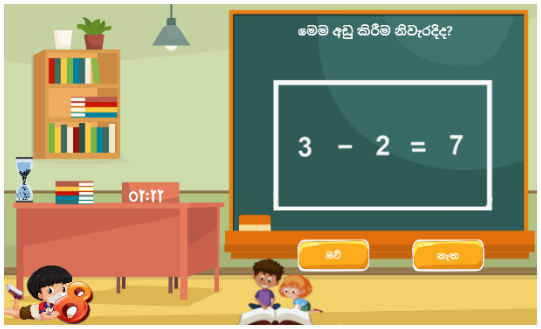
(a) enumerating quiz



(b) comparison of objects



(c) operation: addition



(d) operation: subtraction

Figure 3.5: Operational and Practognostic Dyscalculia screening

In training the model, 225 quiz results were gathered from school children for each stage of the screening test, within the age group 6 to 8. Among them, 75 children are affected by Operational and Practognostic Dyscalculia and 150 children without

Operational and Practognostic Dyscalculia. The trained model was created using the scikit-learn machine learning library in python. A screening test for Dyscalculia was developed which uses a question bank to carry out the test. In the SQLite database, all the questions needed for the screening test are stored. Further, during the screening test, questions are picked randomly in each stage. Then students will get various questions while performing the screening test. Further, quiz result is stored in an SQLite database when a user taking the quiz.

Here, performs classification by finding the hyper-plane that differentiates the two classes by using a Support Vector Machine (SVM) supervised learning algorithm. SVM predicts whether the students or children are having Operational and Practognostic Dyscalculia or not. The Radial Basis Function (RBF) kernel was used along with the Support Vector Machine (SVM) to train the most optimal model for a more accurate result. Because the data set is non-linear as shown in Figure 3.6.

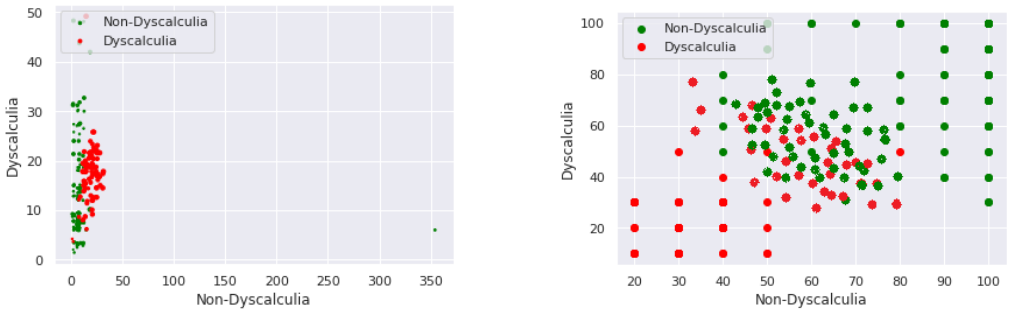


Figure 3.6: Linearly inseparable data set

To screening the risk of having Dyscalculia, the score of the mathematical operation-based quiz covering the addition and subtraction and average time to complete the mathematical operation-based quiz, the score of the enumerating (counting) quiz and an average time to answer the enumerating quiz, and finally sore of the comparison of numbers and objects quiz and an average time to complete the comparison of numbers and objects quiz will be taken as input to the model. These six features fed to the trained SVM model via a python Flask server hosted in the cloud. The SVM model

will then identify the child as regular or Operational and Practognostic Dyscalculia affected child. Then response will be sent to the mobile as shown in Figure 3.7 after clicking the identify button. Further, screening results can be seen by selecting an attempt of the screening report of the student.

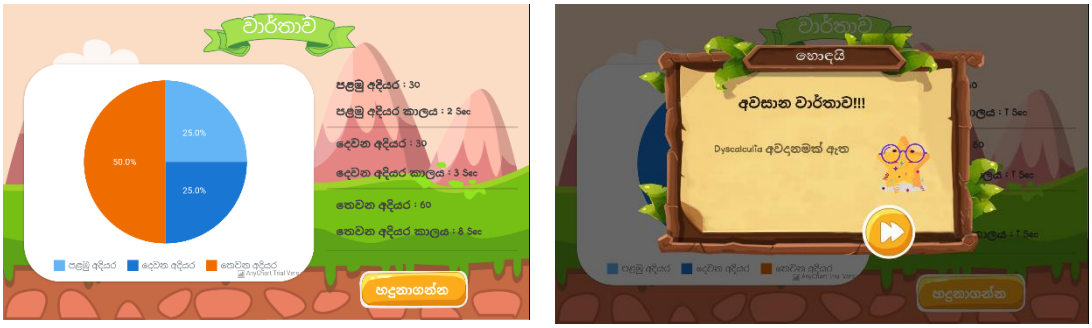
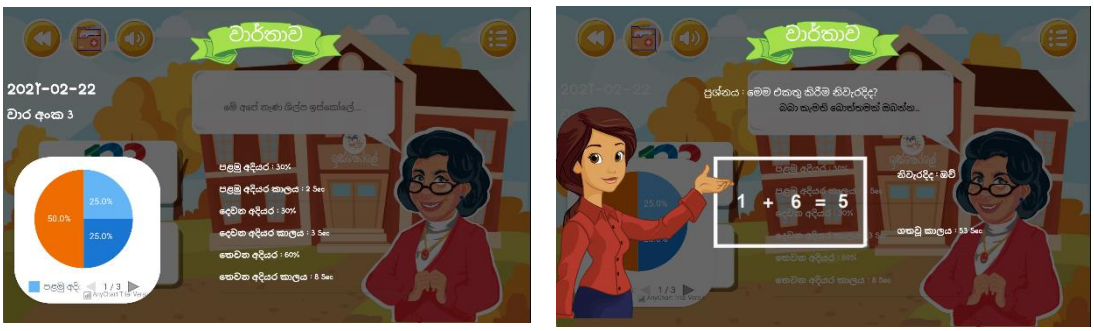


Figure 3.7: Operational and Practognostic Dyscalculia result interface

Since, all the quiz results are stored in SQLite database, previously done quiz results can be shown as reports by a parent or teacher. Figure 3.8.a shows the attempts of each done quiz and Figure 3.8.b shows all the detailed results of the chosen attempt. This will help to parent or teacher to catch the progress of the student.



(a) Attempts of quiz test (b) Details result of each attempt

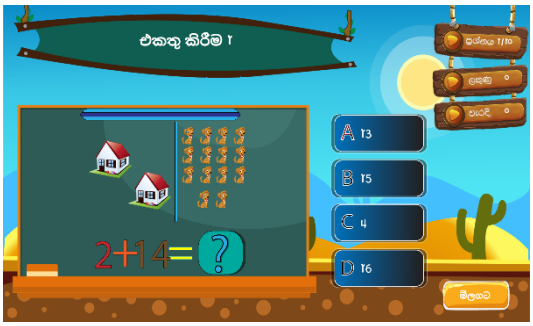
Figure 3.8: Operational and Practognostic Dyscalculia report



(a) Remedy category menu



(b) Remedy instruction



(c) Remedy activity quiz



(d) Remedy quiz review

Figure 3.9: Operational and Practognostic Dyscalculia remedy activities

If the student is at risk of Operational and Practognostic Dyscalculia set of comprehensive activities will be proposed. It was further divided into four categories to practice mathematical addition operation, mathematical subtraction operation, enumerating, and comparison of numbers and objects as shown in Figure 3.9.a. Not only that but also a multisensory learning environment (voice + graphic animation) will be given to students to carry out the practice session as shown in Figure 3.9.b. In mathematical addition operation there are four stages and eventually in each stage students will get the different concept of addition. In the first stage, students will be given a simple exercise to add two numbers and choose the corresponding answer represented in the blue graphical card as shown in Figure 3.9.c. If the student selected the answer correctly the blue graphical card will turn into the green with a congratulation sound. After finishing the practice quiz students can review the quizzes that student has done as shown in Figure 3.9.d. As well as In mathematical subtraction

operations also have four stages and eventually in each stage students will get the different concept of subtraction. Likewise, all the categories have questions to improve the student's skills. After consultation with medical professionals, the remedy was planned to enhance the observation and skills of Operational and Practognostic Dyscalculia children.

3.3.3 Identify Risk of Numeric Dysgraphia

Implemented 'Nana Shilpa' mobile application provides both screening and intervention processes including more other features. For the screening process of this component, it was performed using a number (0 - 9) writing quiz under the direct consultations from medical professionals as shown in Figure 3.10.a and Figure 3.10.b. A drawable canvas is provided by the application to write. The application requires to write digits from 0 to 9 each twice in the provided canvas to reach a more accurate level.



(a) Screening interface 1 (b) Screening interface 2

Figure 3.10: Screening interfaces of ND component

While writing digits, in each attempt, the correctness, erase count, time taken to write the digit as well as boundary touch count are taken per question in order to the Number Dysgraphia prediction. At the end of this task, the results are shown. The results are stored in the SQLite database. The Room Persistence Library (RPL) which is an abstract layer over SQLite is used in order to easily access the backend database through API endpoints. The prediction is shown as a risk of having ND. The most

critical part of this component of this application is the handwriting recognition part. In order to predict the status accurately, that part needs to be done very carefully.

Convolutional Neural Network (CNN) is a deep learning technique which uses for image recognition and processing process especially for pixel data. So, the digit identification part is implemented by using a CNN. A well-trained CNN model is trained to check the accuracy of the written digit. MNIST number dataset which imported from Keras, used by the CNN model. Per each number, it uses 60,000 training image datasets with the dimensions 28 x 28. In the first step, using Keras API, the MNIST number data set is imported. Then comes the processing stage as the second step. Here, it is normalized and resized of the gray-scale image pixel values. Then it is needed 10 output classes since the model can analyze digits between 0 – 9. The main layers conducted digit evaluation process are the input layer, Conv2d layer and the output layer. The input layer is used for inputting the normalized and resized images to the model. Conv2d layer is the one which uses for convolving the dataset image using 32 filters in each size. 10 neurons consist of the output layer for 10 output classes. Over 20 epochs, the model is fit. Nearly, 10,000 image datasets are used as test datasets to evaluate the model. Then the model training phase is done with achieving 99% accuracy. In the CNN, the internal process is done as shown in Figure 3.11.

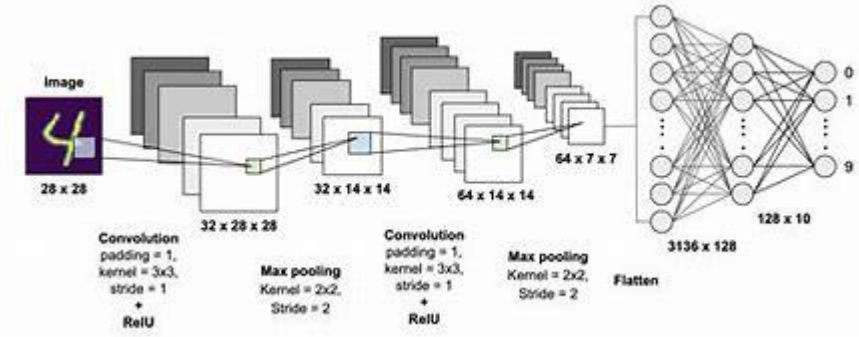


Figure 3.11: Number recognition process: Structure of the CNN

Google Collaboratory which is a free Jupyter Notebook Environment that entirely runs on the cloud is used to train the CNN model. This environment is most compatible when compared to others as Anaconda which is more time consuming and complex since installing phase.

After done the training phase, the trained CNN model is converted to a TensorFlow lite file in order to provide convenient access from the Android environment. TensorFlow is an open-source framework which use to run Machine learning or Deep Learning models to predict analysis. TensorFlow Lite is the lightweight version of TensorFlow that can be used in the Android environment.

After a student writes a digit on the canvas, that written digit is converted to a Bitmap file. The converted CNN model (TensorFlow lite file) is added to the Android environment as an asset file. The mentioned Bitmap file goes through the added asset file in order to process digit recognition. The TensorFlow lite file identifies and then returns the identified number from the Bitmap image. Then the application compares the returned number and the real number which student need to write are equal. If those numbers are equal, that question of the quiz is considered as correct. This is the way the term correctness of the results is processed. It contains whether the written digit is correct or wrong. At the end of the task, the total correctness is shown as how many questions are correct from all questions (from 20 questions) of the quiz.

While doing the quiz, the data are stored in SQLite database. Then data are sent to the cloud-hosted Flask server for the ND prediction. After done the quiz, the results are shown as total correctness, total erase and boundary touch count and the average time taken to write a digit. Those results are used as input datasets to train the Machine Learning model which uses for Numeric Dysgraphia prediction. Random Forest Classifier is used to classify data that has high accuracy compared to algorithms as SVM.

After done the quiz, the application asks to hand over the mobile phone in order to view results as shown in Figure 3.12.a. Results from this process are used to train the ML model. In screening, over 400 students who are in the age range 6 – 8, are used to test the application.



(a) Dialog shown interface. (b) Results shown interface.

Figure 3.12: Number recognition process

After clicking the button to identify as shown in Figure 3.12.b, the Numeric Dysgraphia prediction is shown as a probability as shown in Figure 3.13.

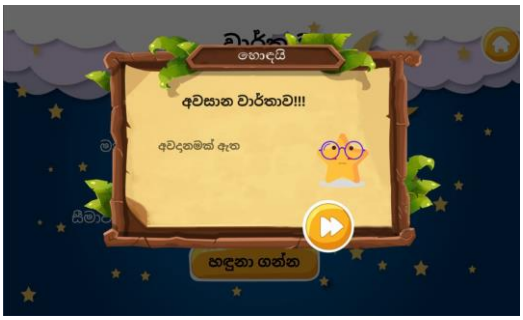
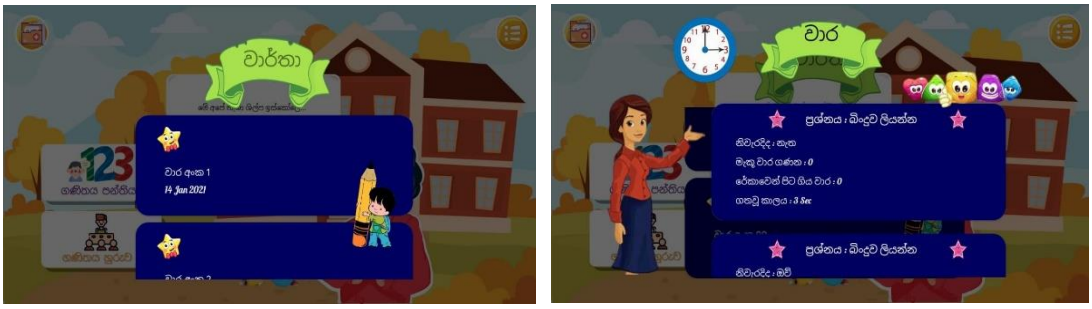


Figure 3.13: ND prediction interface

Since, all the data are stored in SQLite database, previously done quiz results can be shown as reports by a parent or teacher as shown in Figure 3.14.a and Figure 3.14.b. Figure 3.14.a shows the result stages of each done quiz and Figure 3.14.b shows all the detailed results of chosen result stage. The icon for reports is clearly shown in the user interface. This will help to parent or teacher to catch the progress of the student.

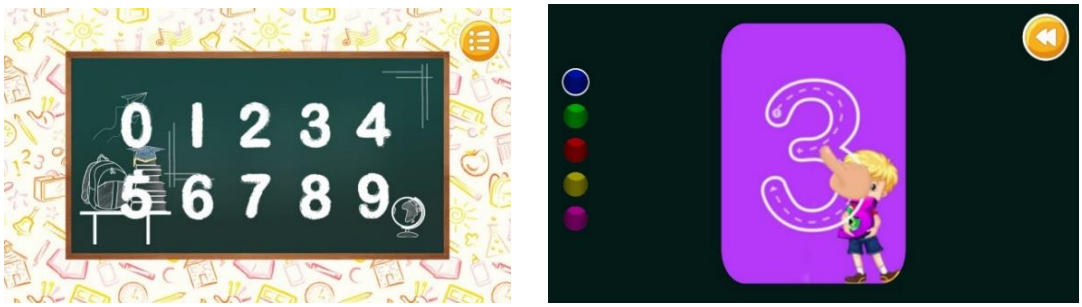
The reports or previously done quiz results can be navigated to the above icon close to the left side on the sub-menu page.



(a) Results in stages (b) Detailed results

Figure 3.14: ND done quiz results shown interface.

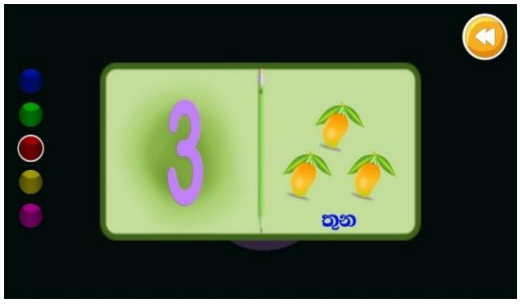
If the application predicts as there is a risk for Numeric Dysgraphia to a student, the intervention also provided by this application based on the guidance of medical and educational professionals. For the purpose of providing a better environment for the intervention, Unity is used. Unity is a cross-platform game engine that provides a component driven design workspace with artist-tool friendly tools. Here, a game-based activity is implemented for the intervention. First, a student can select any preferred number from 0 – 9, as shown in Figure 3.15.a. After selecting a number, then a small demo clip is presented with the guidance to write the number as shown in Figure 3.15.b. Then diagnosed student is prompted to write the number with the guidance as shown in Figure 3.15.c. If only the student writes the number correctly, a pop-up will appear with image appearance for better consideration as shown in Figure 3.15.d.



(a) Intervention digit selection (b) Intervention demo guidance



(c) Intervention digit writing



(d) Intervention pop-up after writing

Figure 3.15: Intervention interfaces in ND component.

Written color can be changed as desired. A diagnosed student needs to practice 6 to 8 months in order to gain a better practice based on the medical professional guidance.

As described, the Numeric Dysgraphia component of this application is implemented. Even though all the guidance and advice were taken from the medical and educational professionals, it is provided with the result as a probability of a risk.

From the loading interface of this application, it is added voice guidance as well as music voice clips when needed in order to develop the application suited for primary school students' mindset.

3.3.4 Identify the Risk of Lexical and Verbal Dyscalculia

Implemented application has the ability to provide screening tests for Lexical & Verbal Dyscalculia and provide remedy activities to overcome mentioned conditions through practices. User can select one type of screening test from a sub-menu as shown in Figure. 3.16.b to perform a screening test and check the probability of having mentioned conditions.



(a) Sub menu (b) Test selection page

Figure 3.16: Navigational menus

If the user goes through Lexical Dyscalculia identifying test, the application will provide voice guidance on how to perform the test. The application will display a random number from a predefined number set as shown in Figure. 3.17.a. Among numbers from 0 to 1000, 12 numbers were taken for considering numbers from three categories with the ones, tens, hundreds four from each category. They are 0, 3, 7, 9, 15, 30, 54, 88, 101, 222, 635, 747. After that user has to press the Microphone button and pronounce the displayed number and press the Next button. Then, the application will provide another number to pronounce. The user needs to perform this test five times and at the end, the application will show the risk of Verbal Dyscalculia as a percentage. The fact consider here is whether the user identifies the displayed number and pronounces it correctly.



(a) Lexical dyscalculia test page (b) Verbal dyscalculia test page

Figure 3.17: Lexical and Verbal Dyscalculia test pages

The application consists of a CNN which is developed using TensorFlow 2.0 and Keras sequential model, to identify Verbal Dyscalculia risk. CNN consists of five layers as shown in Figure 3.18. There are three convolutional layers with batch normalizer, Maxpool2D layers, and “ReLu” activation function in each. The output of them flattens into 1D using a dense layer that consists of 64 neurons and a dropout layer. Then the output feeds into another dense layer with 12 neurons and a softmax classifier used as an activation function. The final output goes through Adam optimizer and sparse categorical cross-entropy used as the loss function for better results.

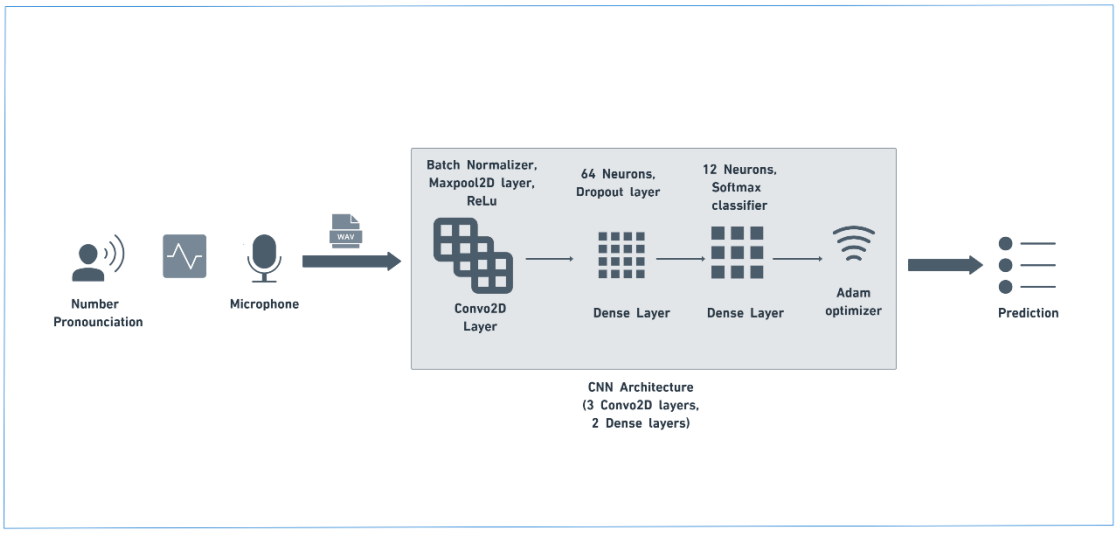


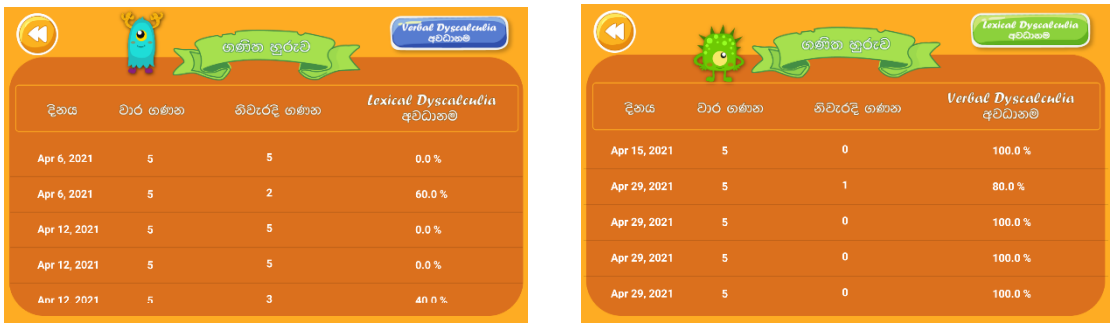
Figure 3.18: CNN architecture of Lexical Dyscalculia prediction

CNN model trained using pronunciations of above-mentioned sample 12 numbers which have taken from primary school student aged between 6 – 8. The audio data set consists of 400 .wav files from each with the length of 3 seconds. The whole data set is split into three parts as train, validation, and test. 0.1 of data set (10% of the whole dataset) taken as test data splits which is used to test the CNN model. 0.1 from remaining data split (9% of the whole dataset) taken as validation split for hyperparameter tuning. The rest of the dataset has taken as the train split. Before feeding it into the CNN model for training, a set of preprocessing steps has to be done. MFCC of the audio file extracted as the feature for audio classification is the best and most commonly used in audio classification. Take snapshots at different segments of

audio and used 13 MFCCs using Librosa audio processing library. Then the MFCCs are stored inside a JSON file as a Python list. Then the list converts into a 3D NumPy array and feeds it into the CNN model for training with 40 epochs and 32 batch sizes. After the training model is stored inside the .h5 file and saved for the prediction process.

Pronunciation of the number taken from the user by the microphone of the mobile as a .wav file. Then it sends to the CNN model for processing through Flask API. After providing an audio file as input, the application will identify whether user pronunciation is correct or wrong. Then the result will send back to the mobile application through the API. Finally, after considering five times of the test the application will display the percentage of Verbal Dyscalculia risk to the user.

If the user chooses the Lexical Dyscalculia identifying test, the application will provide voice guidance on how to perform the Lexical Dyscalculia identifying test. The application will provide a pre-recorded audio of the pronunciation of one number above mentioned at a time. Also, the application will display two numbers including the correct number according to audio as shown in Figure. 3.17.b. The user needs to press the play button and select the correct answer according to the audio from provided two answers. After selecting the correct answer, the user can press the Next button and another number will provide by the application. This test also has to perform for five times and at the end risk of Lexical Dyscalculia will be displayed as a percentage by the application according to the given answers. The fact considering here is whether the user listens to the number pronunciation and selects the correct answer according to it. Lexical Dyscalculia test process through the android application. After the user chooses the answer according to provided voice recording, the application will check whether choose answer is according to the voice record. 12 voice records of the above-selected numbers and correct answers are stored inside the android application. After comparing the given answers of 5 attempts, the application will display the percentage of Lexical Dyscalculia risk to the user.



(a) Lexical Dyscalculia record page (b) Verbal Dyscalculia record page

Figure 3.19: Lexical and Verbal Dyscalculia test record pages

The number of correct answers, date, and the risk percentage of Verbal or Lexical Dyscalculia will be added to the database each time after performing and displaying the identifying test result. Users can refer to these test records as shown in Figure. 3.19 and track the progress of the student at any time.

If the user chooses to go through remedy practices, the application will provide an example activity of how to pronounce & select the correct number as a practice activity. Users can choose from three examples which are given from three categories with the ones, tens, hundreds each. The application will train the user to how to pronounce the number and give some time to pronounce the number and also select the pronounced number from different answers. The remedy activity interface is shown in Figure. 3.20.

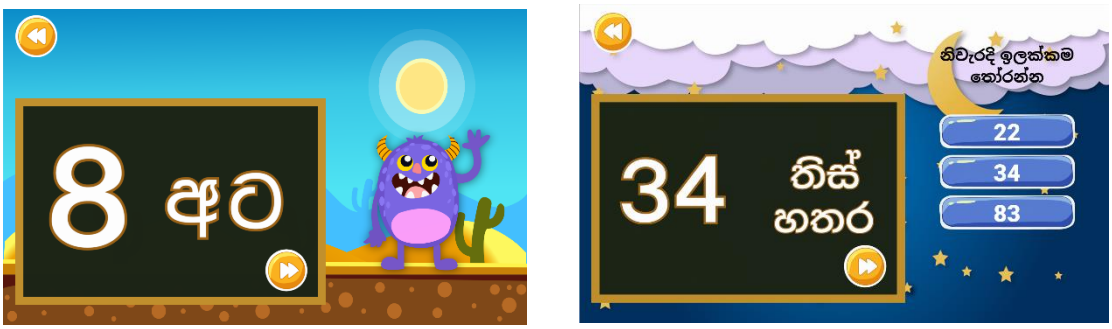


Figure 3.20: Lexical and Verbal Dyscalculia remedy activity pages

3.4 Work Breakdown Structure

The workload of the research project is shown in Appendix C.

3.5 Gantt Chart

The Gantt chart of the research project is shown in Appendix B.

3.6 Commercialization Aspects of the Products

As mentioned earlier, there is a real necessary for low-cost screening and recovery methods regarding these types of conditions. Due to the lack of existence of them, a well-developed and accurate mobile application like “Nana Shilpa” will have high commercial value. The target users of the implemented application are school children from age 6 – 8, their parents, and teachers. Considering the user's conditions and development & deployment cost of the application, the “Nana Shilpa” group planning to offer low cost and free subscription plans for different kind of application users as shown in Table 1.3 and Figure 3.21.

For the individual usage of the application is provided with the cost of Rs.300 and annual subscription costs Rs.600. The schools' plan which covers an age group of six to eight costs Rs. 1500.

Table 3.1: Free & low-cost subscription plans

Subscription Plan	Usage	Cost (Rs.)
1. Monthly subscription	Individual	300
2. Annual subscription	Individual	600
3. Permanent subscription	Individual	1000
4. For schools	One plan for six to eight age group	1500
5. Child orphanage	For all users	Free subscription
6. Low-income families	With proper evidence	Free subscription

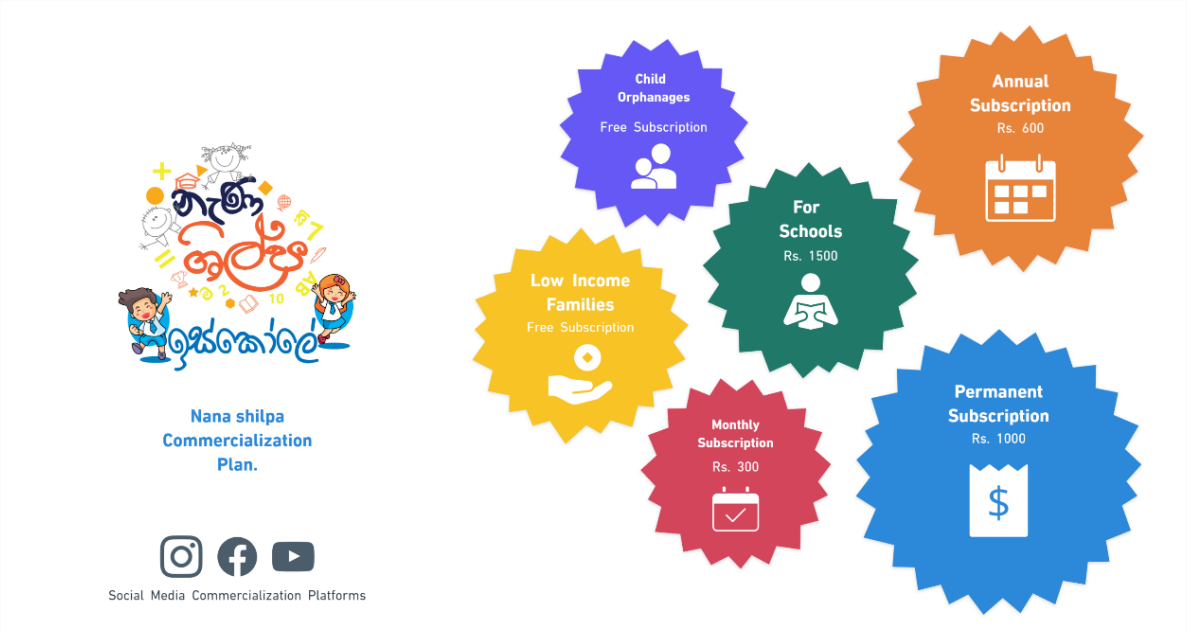


Figure 3.21: Nana Shilpa commercialization plan

Apart from that, we are planning to do social media commercialization regarding the application and its features using popular social media platforms like Instagram, Facebook and YouTube. Also, we are planning to conduct awareness sessions for educating people about these types of conditions as well as the application and its features.

3.7 Testing and Implementation

To ensure both functional and non-functional quality control of the implemented mobile application, a large number of tests have been carried out in both the implementation and testing phase.

3.7.1 Letter level dysgraphia component test cases

Table 3.2: Letter level dysgraphia predict test case (for dysgraphic child)

Test case ID	NST01
Test Case Scenario	Predict the risk of Letter Level Dysgraphia user writes the letters and words incorrectly. (for a child who has a possible risk of having Letter Level Dysgraphia)
Test input data	Handwritten letters and words of a child within age 6 to 10 (.png file)
Test procedure	1. Select “සිංහල ලියන්න” button from main menu. 2. Select “සිංහල පන්තිය” button from letter level dysgraphia sub-menu. 3. Listen to voice guidance on how to perform the test. 4. The user performs the screening test by writing the letters or the word displayed in the upper banner in the canvas. 5. Press “ඉදිරියට යමු” button to go forward and move to the next letter or the word. 6. Perform the process for 8 times. 7. Finish the screening writing test. 8. Press “හරි” button to view the results of the screening test.
Expected output	User should identify as having the risk of Letter Level Dysgraphia user.

Actual output	Identified user as having the risk of Letter Level Dysgraphia user.
Test result	Pass

Table 3.3: Letter level dysgraphia predict test case (for non-dysgraphic child)

Test case ID	NST02
Test Case Scenario	Predict the risk of Letter Level Dysgraphia user writes the letters and words correctly. (for a child who has not a possible risk of having Letter Level Dysgraphia)
Test input data	Handwritten letters and words of a child within age 6 to 10 (.png file)
Test procedure	1. Select "සිංහල ලියන්න" button from main menu. 2. Select "සිංහල පන්තිය" button from the letter level dysgraphia sub-menu. 3. Listen to voice guidance on how to perform the test. 4. The user performs the screening test by writing the letters or the word displayed in the upper banner in the canvas. 5. Press "ඉදිරියට යමු" button to go forward and move to the next letter or the word. 6. Perform the process for 8 times. 7. Finish the screening writing test. 8. Press "හරි" button to view the results of the screening test.
Expected output	User should identify as not having the risk of Letter Level Dysgraphia user.
Actual output	Identified user as not having the risk of Letter Level Dysgraphia user.
Test result	Pass

3.7.2 Operational & practognostic dyscalculia test cases

Table 3.4: Dyscalculia predict test case (for normal child)

Test case ID	NST03
Test Case Scenario	Predict Operational and Practognostic Dyscalculia – User has done mathematical quiz accurately (for normal child)
Test input data	Mathematical quiz results (90, 4.32, 100, 2.54, 90, 7.35)
Test procedure	1. Select the Math problem (ගණිත ගැටලු) button in the main menu. 2. Select the Math class (ගණිත පන්තිය) button in the sub-menu to do the math quiz. 3. Proceed with the enumerating quiz as stage 1 and click the next button in the dialog box after completion. 4. Proceed with the comparison objects and numbers quiz as stage 2 and click the next button in the dialog box after completion. 5. Proceed with the mathematical operation-based quiz covering the addition and subtraction quiz as stage 3 and click the next button in the dialog box after completion. 6. Display the results of each stage and average time to answer each question on the result page. 7. Click predict (හඳුනාගන්න) button. 8. Display whether children who suffer from dyscalculia or non- dyscalculia (සමත් or අසමත්).

Expected output	The user Should be notified as a non- dyscalculic child (සමත්).
Actual output	The user is notified as a non- dyscalculia child (සමත්).
Test result	Pass

Table 3.5: Dyscalculia predict test case (for dyscalculic child)

Test case ID	NST04
Test Case Scenario	Predict Operational and Practognostic Dyscalculia – User has done mathematical quiz weakly (for dyscalculic child)
Test input data	Mathematical quiz results (40, 11.42, 50, 8.56, 30, 14.59)
Test procedure	1. Select the Math problem (ගණිත ගැටලු) button in the main menu. 2. Select the Math class (ගණිත පන්තිය) button in the sub-menu to do the math quiz. 3. Proceed with the enumerating quiz as stage 1 and click the next button in the dialog box after completion. 4. Proceed with the comparison objects and numbers quiz as stage 2 and click the next button in the dialog box after completion. 5. Proceed with the mathematical operation-based quiz covering the addition and subtraction quiz as stage 3 and click the next button in the dialog box after completion. 6. Display the results of each stage and average time to answer each question on the result page. 7. Click predict (හඳුනාගන්න) button. 8. Display whether children who suffer from Dyscalculia or non- Dyscalculia (සමත් or අසමත්).

Expected output	The user Should be notified as a dyscalculic child (අසමත්).
Actual output	The user is notified as a dyscalculic child (අසමත්).
Test result	Pass

Table 3.6: Dyscalculia intervention test case for addition operation success

Test case ID	NST05
Test Case Scenario	Attempt remedy activity for Operational and Practognostic Dyscalculia to improve the condition – User done addition operation quiz level 1 with 100% marks (success).
Test input data	Addition operation quiz level 1 results.
Test procedure	1. Select the Math problem (ගණිත ගැටලු) button in the main menu. 2. Select the Math class (ගණිත හුරුව) button in the submenu to do the math quiz. 3. Select the addition operation (එකතු කිරීම) category from the sub-menu of the remedy. 4. Select level 1 from the addition levels. 5. Listen to the instruction to do the math quiz. 6. Click the next button to start level 1 of the addition quiz. 7. Perform the quiz. 8. After completing the test redirect to the result page and display the quiz result.
Expected output	Should unlock the next level when the user gets 100% marks.
Actual output	Unlock the next level
Test result	Pass

Table 3.7: Dyscalculia intervention test case for addition fail

Test case ID	NST06
Test Case Scenario	Remedy activity for Operational and Practognostic Dyscalculia to improve the condition – User has done addition operation quiz level 1 with 60% marks (fails).
Test input data	Addition operation quiz level 1 results.
Test procedure	1. Select the Math problem (ගණිත ගැටලු) button in the main menu. 2. Select the Math class (ගණිත හුරුව) button in the sub-menu to do the math quiz. 3. Select the addition operation (එකතු කිරීම) category from the sub-menu of the remedy. 4. Select level 1 from the addition levels. 5. Listen to the instruction to do the math quiz. 6. Click the next button to start level 1 of the addition quiz. 7. Perform the quiz by selecting the answers. 8. After completing the test redirect to the result page and display the quiz result.
Expected output	Should not unlock the next level when the user gets below 100% marks.
Actual output	Not unlock the next level
Test result	Pass

Table 3.8: Dyscalculia intervention test case for subtracting operation success

Test case ID	NST07
Test Case Scenario	Attempt remedy activity for Operational and Practognostic Dyscalculia to improve the condition – User has done subtract operation quiz level 1 with 100% marks.
Test input data	Subtract operation quiz level 1 results.
Test procedure	1. Select the Math problem (ගණිත ගැටලු) button in the main menu. 2. Select the Math class (ගණිත හුරුව) button in the sub-menu to do the math quiz. 3. Select the subtract operation (අඩු කිරීම) category from the sub-menu of the remedy. 4. Select level 1 from the subtract levels. 5. Listen to the instruction to do the subtract math quiz. 6. Click the next button to start level 1 of the subtract quiz. 7. Perform the subtract quiz by selecting the answers. 8. After completing the test redirect to the result page and display the quiz result.
Expected output	Should unlock the next level when the user gets 100% marks.
Actual output	Unlock the next level
Test result	Pass

Table 3.9: Dyscalculia intervention test case for subtract operation fail

Test case ID	NST08
Test Case Scenario	Attempt remedy activity for Operational and Practognostic Dyscalculia to improve the condition – User has done subtract operation quiz level 1 with 80% marks.
Test input data	Subtract operation quiz level 1 results.
Test procedure	1. Select the Math problem (ගණිත ගැටලු) button in the main menu. 2. Select the Math class (ගණිත හුරුව) button in the sub-menu to do the math quiz. 3. Select the subtract operation (අඩු කිරීම) category from the sub-menu of the remedy. 4. Select level 1 from the subtract levels. 5. Listen to the instruction to do the subtract math quiz. 6. Click the next button to start level 1 of the subtract quiz. 7. Perform the subtract quiz by selecting the answers. 8. After completing the test redirect to the result page and display the quiz result.
Expected output	Should not unlock the next level when the user gets below 100% marks.
Actual output	Not unlock the next level
Test result	Pass

3.7.3 Numeric dysgraphia test cases

Table 3.10: ND test case for screening test of a normal student

Test Case ID	NST09
Test Case Scenario	Screening for Numeric Dysgraphia – for normal primary school student.
Test Input Data	Bitmap version of written numbers by the student
Test Procedure	1. Selects the button for writing math (ගණිතය ලියන්න) in the main menu. 2. Selects the button for math class (ගණිත පන්තිය) in the sub-main menu. 3. Starts to do the activity by writing numbers from 0 to 9. 4. The app displays the results of the done quiz. 5. Clicks the button (හදුනාගන්න) to predict the Numeric dysgraphia situation. 6. The app displays the results based on quiz results.
Expected Outcome	The app should display as non-dysgraphic situation.
Actual Outcome	App displays as non-dysgraphic situation.
Test Result	Pass

Table 3.11: ND test case for screening test of dysgraphic student

Test Case ID	NST10
Test Case Scenario	Screening for Numeric Dysgraphia – for dysgraphic primary school student.
Test Input Data	Bitmap version of written numbers by the student
Test Procedure	1. Selects the button for writing math (ගණිතය ලියන්න) in the main menu. 2. Selects the button for math class (ගණිත පන්තිය) in the sub-main menu. 3. Starts to do the activity by writing numbers from 0 to 9. 4. The app displays the results of the done quiz. 5. Clicks the button (හඳුනාගන්න) to predict the Numeric dysgraphia situation. 6. The app displays the results based on quiz results.
Expected Outcome	The app should display as dysgraphic situation.
Actual Outcome	App displays as dysgraphic situation.
Test Result	Pass

Table 3.12: ND test case for correctly done intervention process.

Test Case ID	NST11
Test Case Scenario	Correctly done the intervention process.
Test Input Data	User written digit
Test Procedure	1. Selects the button for writing math (ගණිතය ලියන්න) in the main menu. 2. Selects the button for math class (ගණිත හුරුව) in the sub-main menu. 3. The system shows a number set from 0 to 9. 4. The user chooses any preferred number from clicking on the number. 5. The system provides guidance using a demo video clip. 6. The user writes on the number correctly based on the guidance.
Expected Outcome	A small pop-up has to appear with the image appearance.
Actual Outcome	A pop-up has appeared.
Test Result	Pass

Table 3.13: ND test case for the incorrectly done intervention process.

Test Case ID	NST12
Test Case Scenario	Incorrectly done the intervention process.
Test Input Data	User written digit
Test Procedure	1. Selects the button for writing math (ගණිතය ලියන්න) in the main menu. 2. Selects the button for math class (ගණිත හුරුව) in the sub-main menu. 3. The system shows a number set from 0 to 9 in the number selection interface. 4. The user chooses any preferred number from clicking on the number. 5. The system provides previous guidance using a demo video clip. 6. The user writes on the number incorrectly based on the guidance.
Expected Outcome	A pop-up should not appear, and the system should expect to write the number again.
Actual Outcome	A pop-up has not appeared, and the system expects to write the number again.
Test Result	Pass

3.7.4 Lexical and verbal dyscalculia test cases

Table 3.14: Lexical Dyscalculia predict test case (for lexical dyscalculic child)

Test Case ID	NST13
Test Case Scenario	Predict the risk of Lexical Dyscalculia – User pronounces number incorrectly.
Test Input Data	Audio input (.wav)
Test Procedure	1. Select “ගණිත හුරුව” button on the main menu. 2. Select “ගණිත පන්තිය” button from 1st sub-menu. 3. Select “උත්තර කියවන්න” button from 2nd sub-menu. 4. Listen to voice guidance on how to perform the test. 5. Press the Mic button and pronounce the displayed latter. 6. Press the Next button to perform the test again. 7. Perform test five times. 8. Press the Results button to view previous records.
Expected Outcome	The user should identify as a Lexical Dyscalculic user. (Risk above 50% accordingly)
Actual Outcome	Identified user as a Lexical Dyscalculic user. (Risk above 50% accordingly)
Test Result	Pass

Table 3.15: Lexical Dyscalculia predict test case (for non-lexical dyscalculic child)

Test Case ID	NST14
Test Case Scenario	Predict the risk of Lexical Dyscalculia – User pronounces number correctly.
Test Input Data	Audio input (.wav)
Test Procedure	1. Select “ගණිත හුරුව” button on the main menu. 2. Select “ගණිත පන්තිය” button from 1st sub-menu. 3. Select “උත්තර කියවන්න” button from 2nd sub-menu. 4. Listen to voice guidance on how to perform the test. 5. Press the Mic button and pronounce the displayed latter. 6. Press the Next button to perform the test again. 7. Perform test five times. 8. Press the Results button to view previous records.
Expected Outcome	The user should identify as a Non- Lexical Dyscalculic user. (Risk less than 50% accordingly)
Actual Outcome	The user identified as a Non- Lexical Dyscalculic user. (Risk less than 50%)
Test Result	Pass

Table 3.16: Verbal Dyscalculia predict test case (for non-verbal dyscalculic child)

Test Case ID	NST15
Test Case Scenario	Predict the risk of Verbal Dyscalculia – User chooses correct answer.
Test Input Data	Quiz answers
Test Procedure	1. Select “ගණිත හුරුව” button on the main menu. 2. Select “ගණිත පන්තිය” button from 1st sub-menu. 3. Select “උත්තර තෝරන්න” button from 2nd sub-menu. 4. Listen to voice guidance on how to perform the test. 5. Press the Play button and listen to the voice clip provided by the application. 6. Select the correct number according to the voice clip. 7. Press the Next button to perform the test again. 8. Perform test five times. 9. Press the Results button to view previous records.
Expected Outcome	The user should identify as a Non-verbal Dyscalculic user. (Risk less than 50% accordingly)
Actual Outcome	The user identified as a Non-verbal Dyscalculic user. (Risk less than 50%)
Test Result	Pass

Table 3.17: Verbal Dyscalculia predict test case (for verbal dyscalculic child)

Test Case ID	NST16
Test Case Scenario	Predict the risk of Verbal Dyscalculia – User chooses the wrong answer.
Test Input Data	Quiz answers
Test Procedure	1. Select “ගණිත හුරුව” button on the main menu. 2. Select “ගණිත පන්තිය” button from 1st sub-menu. 3. Select “උත්තර තෝරන්න” button from 2nd sub-menu. 4. Listen to voice guidance on how to perform the test. 5. Press the Play button and listen to the voice clip provided by the application. 6. Select the correct number according to the voice clip. 7. Press the Next button to perform the test again. 8. Perform test five times. 9. Press the Results button to view previous records.
Expected Outcome	The user should identify as a Verbal Dyscalculic user. (Risk less than 50% accordingly)
Actual Outcome	The user identified as a Verbal Dyscalculic user. (Risk less than 50%)
Test Result	Pass

4 RESULT AND DISCUSSION

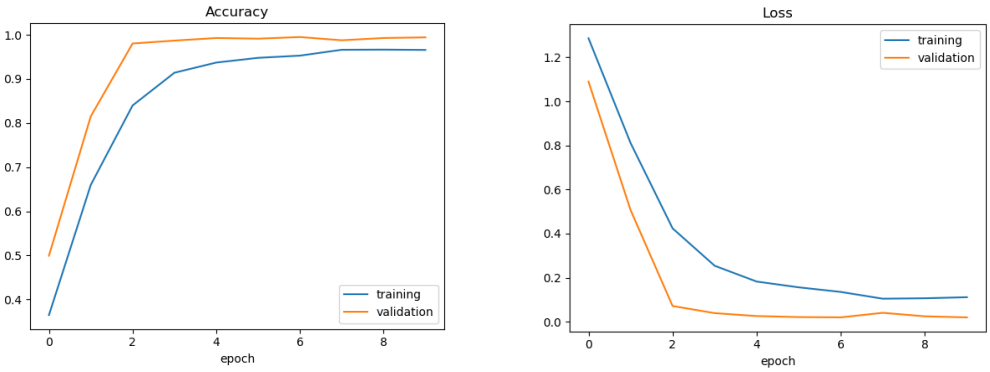
4.1 Results

‘Nana Shilpa’ has been developed for screening and refinement system to identify Dysgraphia and Dyscalculia in Sri Lankan primary school students around different regions across Sri Lanka. The mobile application is implemented using android, SQLite database, Flask and Tensor flow as the middleware technology and the backend of the application is python. The research has been used machine learning concepts and algorithms, image processing, voice recognition techniques and frequency comparison in the implementation.

In all four components, ‘Nana Shilpa’ had obtained over 97% of accuracy levels which can grantee the positive contribution on screening the risk of Dysgraphia and Dyscalculia disability conditions.

4.1.1 Letter level dysgraphia identification

More than 1000 letters were collected for each word and letter from primary school students and preprocessing using gray-scale conversion, noise reduction and histogram and segmentation. Convolutional Neural Network (CNN) has been chosen since it is designed to map 2D or multi-dimensional image data to one-dimensional output variable. Also, CNN delivers high accuracy for the image classification and recognition process. After preprocessing, the data was fed to the CNN model in order to identify the accuracy of the letter or word.



(a) CNN model accuracy (b) CNN model loss graph

Figure 4.1: CNN model accuracy graphs for letter level dysgraphia screening

```
Epoch 2/10
109/109 [==============================] - 16s 145ms/step - loss: 0.8111 - accuracy: 0.6597 - val_loss: 0.5072 - val_accuracy: 0.8151
Epoch 3/10
109/109 [==============================] - 14s 131ms/step - loss: 0.4237 - accuracy: 0.8398 - val_loss: 0.0716 - val_accuracy: 0.9803
Epoch 4/10
109/109 [==============================] - 15s 136ms/step - loss: 0.2542 - accuracy: 0.9140 - val_loss: 0.0392 - val_accuracy: 0.9868
Epoch 5/10
109/109 [==============================] - 15s 137ms/step - loss: 0.1824 - accuracy: 0.9371 - val_loss: 0.0260 - val_accuracy: 0.9927
Epoch 6/10
109/109 [==============================] - 15s 141ms/step - loss: 0.1563 - accuracy: 0.9478 - val_loss: 0.0213 - val_accuracy: 0.9912
Epoch 7/10
109/109 [==============================] - 16s 143ms/step - loss: 0.1354 - accuracy: 0.9528 - val_loss: 0.0200 - val_accuracy: 0.9949
Epoch 8/10
109/109 [==============================] - 16s 144ms/step - loss: 0.1047 - accuracy: 0.9660 - val_loss: 0.0407 - val_accuracy: 0.9876
Epoch 9/10
109/109 [==============================] - 16s 143ms/step - loss: 0.1067 - accuracy: 0.9664 - val_loss: 0.0247 - val_accuracy: 0.9927
Epoch 10/10
109/109 [==============================] - 16s 147ms/step - loss: 0.1115 - accuracy: 0.9657 - val_loss: 0.0200 - val_accuracy: 0.9942
Test Score =  0.02710871584713459
Test Accuracy = 0.9929783344268799

Process finished with exit code 0
```

Figure 4.2: Model accuracy report for CNN model

CNN model which had used for the screening process of letter level Dysgraphia had obtained high accuracy level due to the supplementary database used in the training process. After the model training, as shown in Figure 4.2, the CNN model had obtained over 98% of accuracy for the test input data and Figure 4.1 shows the training and validation accuracy against to 8 epochs of the model.

4.1.2 Operational and practognostic dyscalculia identification

In this study, 225 quiz results were gathered from school children for each stage of the screening test. Finally, collected data were fed into the SVM model as training data. The SVM classification algorithm was trained using above mentioned six features to predict whether the children have Dyscalculia or not. Nearly 98% accuracy was achieved identification of Operational and Practognostic Dyscalculia. Among the 45 testing instances, 44 instances correctly classified which is 97.78% accuracy of the testing, and only 1 instance was predicted as incorrectly. Figure 4.3 shows the SVM training set confusion matrix. It shows the estimated value against the actual value by the classifier.

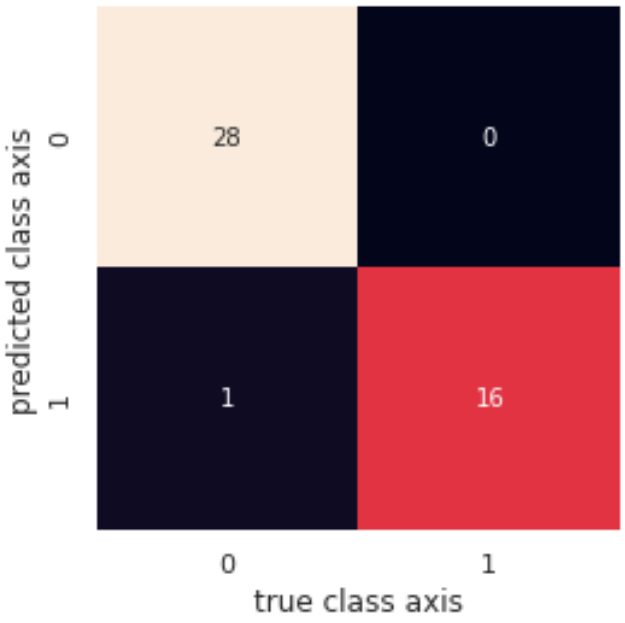
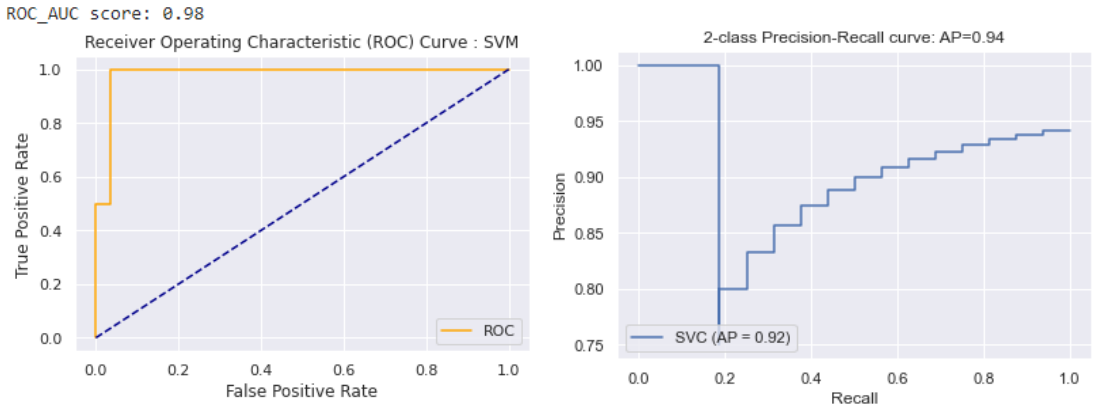


Figure 4.3: Confusion matrix for SVM

SVM gives high precision and recall in the above-mentioned matrix and has been obtained at a weighted average accuracy of 0.98 and with a weighted mean recall of 0.98 showing that the classifier returns an accurate result as shown in Figure 4.4.b, and the curve of ROC describes the rate of true positive values against false positive values in the range of possible threshold values as shown in Figure 4.4.a. The classification report was created to display the most important classification metrics as shown in Figure 4.5.



a) ROC curve for Dyscalculia identification

(b) Precision recall curve for Dyscalculia

Figure 4.4: Precision recall and ROC curve for Dyscalculia screening

Classification Report :

	precision	recall	f1-score	support
No Dyscalculia	0.97	1.00	0.98	28
Dyscalculia	1.00	0.94	0.97	17
accuracy			0.98	45
macro avg	0.98	0.97	0.98	45
weighted avg	0.98	0.98	0.98	45

Figure 4.5: Classification report for SVM

4.1.3 Numeric dysgraphia identification

Here, based on the results presented from the CNN model trained in the Google Collaboratory environment, the imported MNIST number data from Keras has shown a nearly 99% accuracy in the digit recognition process as shown in Figure 4.6.

```
]: experiment_accuracy = sm.accuracy_score(y_test, prediction_y)
   print('Accuracy score is : ', str(experiment_accuracy))

   Accuracy score is :  0.9896907216494846
```

Figure 4.6: ND CNN model accuracy evidence.

The number model accuracy is shown in a diagram as shown in below Figure 4.7. Here the diagram shows the accuracy against the epochs.

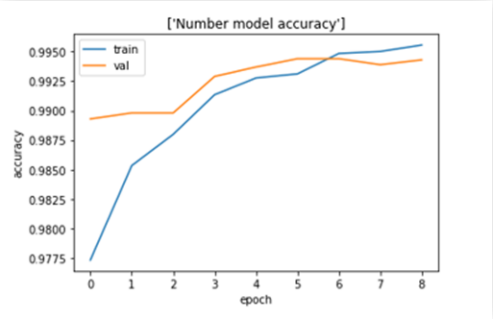
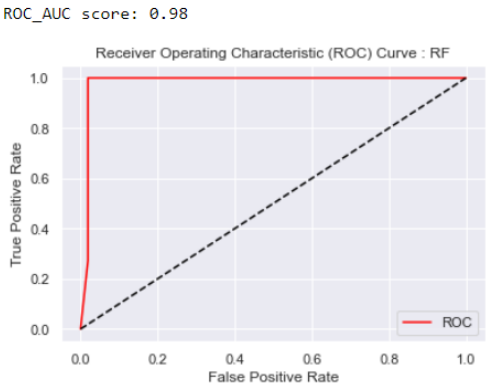


Figure 4.7: Number model accuracy

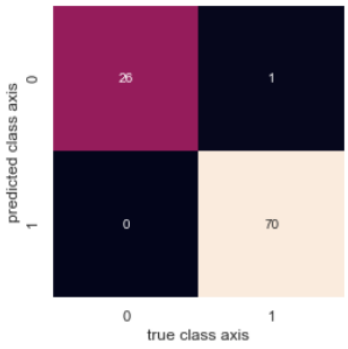
When it comes to the trained ML model in the Jupytor notebook environment, for the Numeric Dysgraphia component, it is used five algorithms to check what is the best algorithm for the prediction. So, the below factors show the used algorithm and the received accuracy achieved while training.

- Random Forest Algorithm — Accuracy of 99% (98.99%)
- Decision Tree — Accuracy of 91%
- Support Vector Machine Algorithm — Accuracy of 90%
- Logistic Regression — Accuracy of 88%
- K-Nearest Neighbors — Accuracy of 89%

So, from the accuracy percentages shown here, for the Numeric Dysgraphia prediction, the Random Forest algorithm is chosen since it shows a high accuracy rate. As shown in Figure 4.8.a and 4.8.b, those diagrams show the ROC curve for ND identification and Confusion matrix for RF respectively.



(a) ROC curve for ND identification



(b) Confusion matrix for RF

Classification report:

	precision	recall	f1-score	support
Numeric Dysgraphia	0.98	1.00	0.99	51
No Numeric Dysgraphia	1.00	0.97	0.99	34
accuracy			0.99	85
macro avg	0.99	0.99	0.99	85
weighted avg	0.99	0.99	0.99	85

(c) Classification report for the ND using RF

Figure 4.8: Results from the trained ML model using RF

As shown in Figure 4.8.c, it is shown all the results from the trained ML model. In Figure 4.8.c, the classification report was created to represent the main metrics of classifications. Here, precision, recall, f1-score and support all are characteristics of a classifier. Precision represents the ability to not provide a positive instance as negative. Then, recall means the ability to get all positive values. F1-score represents the precision and recall's weighted mean. 1.00 is the best case and 0.00 is the worst case. By contrasting and comparing with other algorithms' classification reports, it can come to a conclusion as Random Forest is the best algorithm for ND identification since it contains a high precision value, high recall value and high F1-score value.

4.1.4 Verbal dyscalculia identification

Initially, selected 12 numbers for the screening process by considering ones, tens, hundreds. Data needed for the CNN training process gathered from primary school students aged between 6 – 8. The audio data set consists of 400 .wav files from each with a length of 3 seconds.

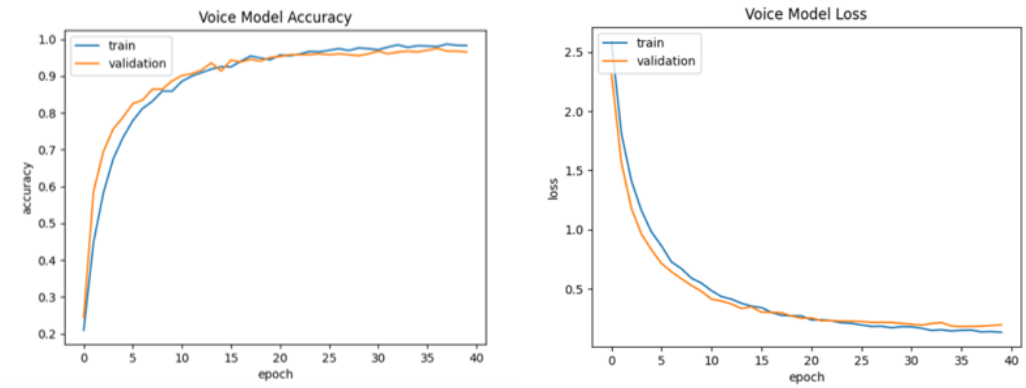


Figure 4.9: CNN model accuracy & loss graphs

As shown in Figure 4.9 the sequential Keras CNN model accrued 95% of test accuracy and 0.19 test error in the training process with 40 epochs.

4.2 Research Findings

The main four components of this implemented ‘Nana Shilpa’ are for mathematical problems solving based on basic mathematical concepts, voice recognition for numbers, handwritten letter recognition and handwritten number recognition. From these components, voice recognition for numbers, handwritten letter recognition and handwritten number recognition use the CNN model for the recognition process. CNN is more powerful and accurate in data classification problems when compared to other models such as Artificial Neural Networks (ANN). And also, CNN uses more data in the training phase. When use more data to feed the CNN, it can be achieved higher accuracy. So, the CNN model was the best option for voice, handwritten number and handwritten letter recognition.

Table 4.1: Achieved CNN model accuracy on components.

Component	Achieved Accuracy using CNN model
Lexical and Verbal Dyscalculia	96%
Letter Dysgraphia	99%
Numeric Dysgraphia	99%

In the prediction stage in components for mathematical problems solving based on basic mathematical concepts and digit recognition, ML models were trained. In predicting Operational and Practognostic Dyscalculia, it is achieved 98% accuracy using the SVM algorithm in the ML model. Overall, all the algorithms predicted over 85% for the Dyscalculia prediction. In Numeric Dysgraphia prediction, as the high accuracy achieved algorithm, RF is chosen as the best for digit recognition. It has achieved 99% accuracy. So, RF is shown as the most accurate classifier for digit recognition while SVM is turned to be the most accurate classifier for mathematical problems solving component. High precision value, recall value and F1-score have been reached by using these mentioned algorithms in respective components.

Table 4.2: Achieved ML model accuracy based on six different algorithms.

Component	SVM	RF	KNN	Logistic Regression	Decision Tree	Naïve Bias
Accuracy of predicting Operational and Practognostic Dyscalculia	98%	91%	NT	NT	90%	88%
Accuracy of predicting Numeric Dysgraphia Accuracy	90%	99%	89%	88%	91%	NT

4.3 Discussion

Implemented 'Nana Shilpa' mobile application contains main four components which are Verbal and Lexical Dyscalculia, Operational and Practognostic Dyscalculia, Letter Level Dysgraphia and Numeric Dysgraphia. These are the categories where most of the students of LDs suffering. When implementing this application, it is tested many algorithms to check what algorithm is best suited for the prediction and then select the algorithm which achieved high accuracy in the ML model training phase. In the testing phase, all the test cases show that the application works well as expected. Both functional and non-functional testing was done in order to provide a better product. When examined and analyzed a lot of past researches, in Sri Lanka, there is not any application or solution developed to screening these mentioned learning disabilities and provide intervention to them. It is used in the native language in the application in order to provide successful guidance throughout the application. Attractive theme-based user interfaces are designed for the purpose of keeping the direct attention of a student. So, implemented the 'Nana Shilpa' mobile application will provide better support for the students who suffer from mentioned learning disabilities.

As future work, we are planning to expand this application to different age groups, different language options and also different learning disabilities not used in this application. Apart from this, the theme, used colors, button, text, voice guidance like features in the application will be customized in a different attractive manner. We will hope to provide a better solution for Learning Disability prediction with higher accuracy and intervention process in advance.

5 SUMMARY OF EACH STUDENT'S CONTRIBUTION

Table 5.1: Summary of each student contribution

Member	Component	Contribution
1. Hewapathirana C.D.D.	Identify the risk for Operational and Practognostic Dyscalculia and provide activities to improve the condition.	❖ Collect mathematical quiz result from the school. ❖ Design UX and UI for the Dyscalculia component. ❖ Implementing Operational and Practognostic Dyscalculia screening test using the enumerating (counting), comparison of numbers and objects, and mathematical operation-based quiz. ❖ Create a classification model for predict Dyscalculia using the SVM along with the RBF kernel. ❖ Generate reports for screening process and keep track of the progress of the dyscalculic children for future review. ❖ Implement remedy activities to improve the condition of Dyscalculia children. ❖ Implement a multisensory environment for Operational & Practognostic Dyscalculia remedy activity.
2. Abeysinghe A.D.K.S.	Identify the risk of Letter level Dysgraphia and provide activities to improve the condition.	❖ Handwritten Sinhala letter and word collection. ❖ Design UX and UI for the Letter level Dysgraphia component. ❖ Implement the screening test process to identify the risk for Letter level Dysgraphia. ❖ Create and train the CNN model to predict the Dysgraphia condition using handwritten letters and words.

		❖ Report generation for screening test results and keep a record of Letter Level Dysgraphia condition for further review. ❖ Implement interactive intervention practice activities to improve the risk condition.
3. Maheshani H.M.P.	Identify the risk for Numeric Dysgraphia and provide activities to improve the condition.	❖ Design UX and UI for the Numeric Dysgraphia component. ❖ Implement the screening test activity to identify the risk for Numeric Dysgraphia. ❖ Train the CNN model to process the digit identification. ❖ Train the ML model using RF classifier to predict the risk for Numeric Dysgraphia. ❖ Keep the track of progress by storing and showing the previously done quiz results. ❖ Develop the intervention process to improve the condition.
4. Liyanage T.J.P.S.	Identify the risk of Lexical and Verbal Dyscalculia and provide activities to improve the condition	❖ Collect voice data to implement the CNN model. ❖ Design UX and UI of the application. ❖ Provide intervention tests to identify the risk of Lexical Dyscalculia. ❖ Provide intervention tests to identify the risk of Verbal Dyscalculia. ❖ Provide remedy activities for identified users with a risk. ❖ Generate test reports and store them for further review. ❖ Train a CNN model to predict the risk of Lexical Dyscalculia.

6 CONCLUSION

Worldwide, there are many solutions were designed based on Learning disabilities, but in Sri Lanka, it's very hard to find conducted researches as well as solutions in mentioned areas. So, through this conducted research, it was implemented a mobile-based application in the native language (Sinhala) for the screening and refinement processes for the Dyscalculia and Dysgraphia Learning Disabilities for the primary school students who are in the age range 6 – 8. This application predicts the risk as a probability of having Dyscalculia or Dysgraphia with highly accurate results. If the result is predicted as there is a probability of having mentioned learning disabilities, the intervention also provides by the application.

Since this implemented application is based on native language (Sinhala) and on Android platform, eventually, we have planned to explore it to multiple languages, different mobile platform like iOS as well as a solution which will be web-based also. Furthermore, the application will be expanded for different age groups while improving datasets from those groups. As planned future work, we will hope to provide a better solution which fulfils these requirements.

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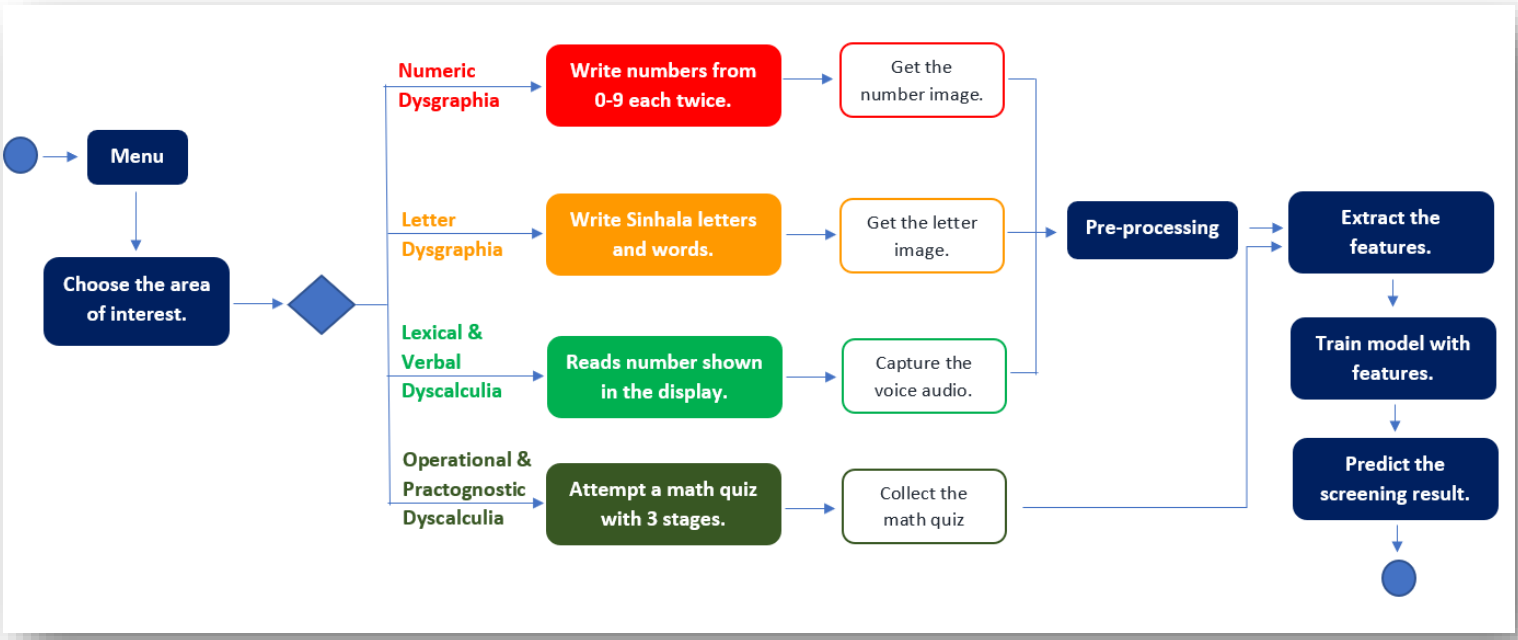
APPENDICES

Appendix - A: Application Logo (Nana Shilpa)

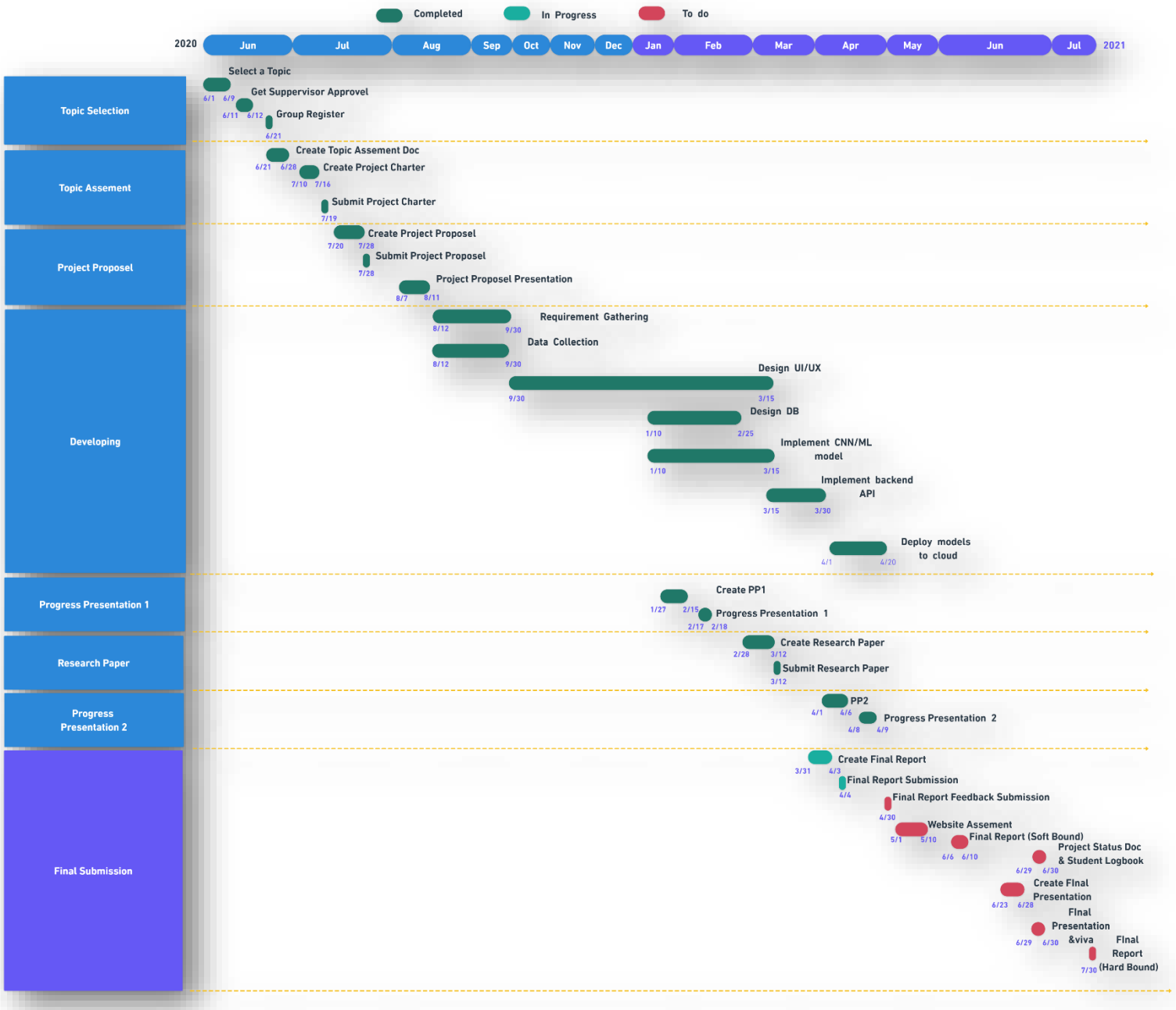




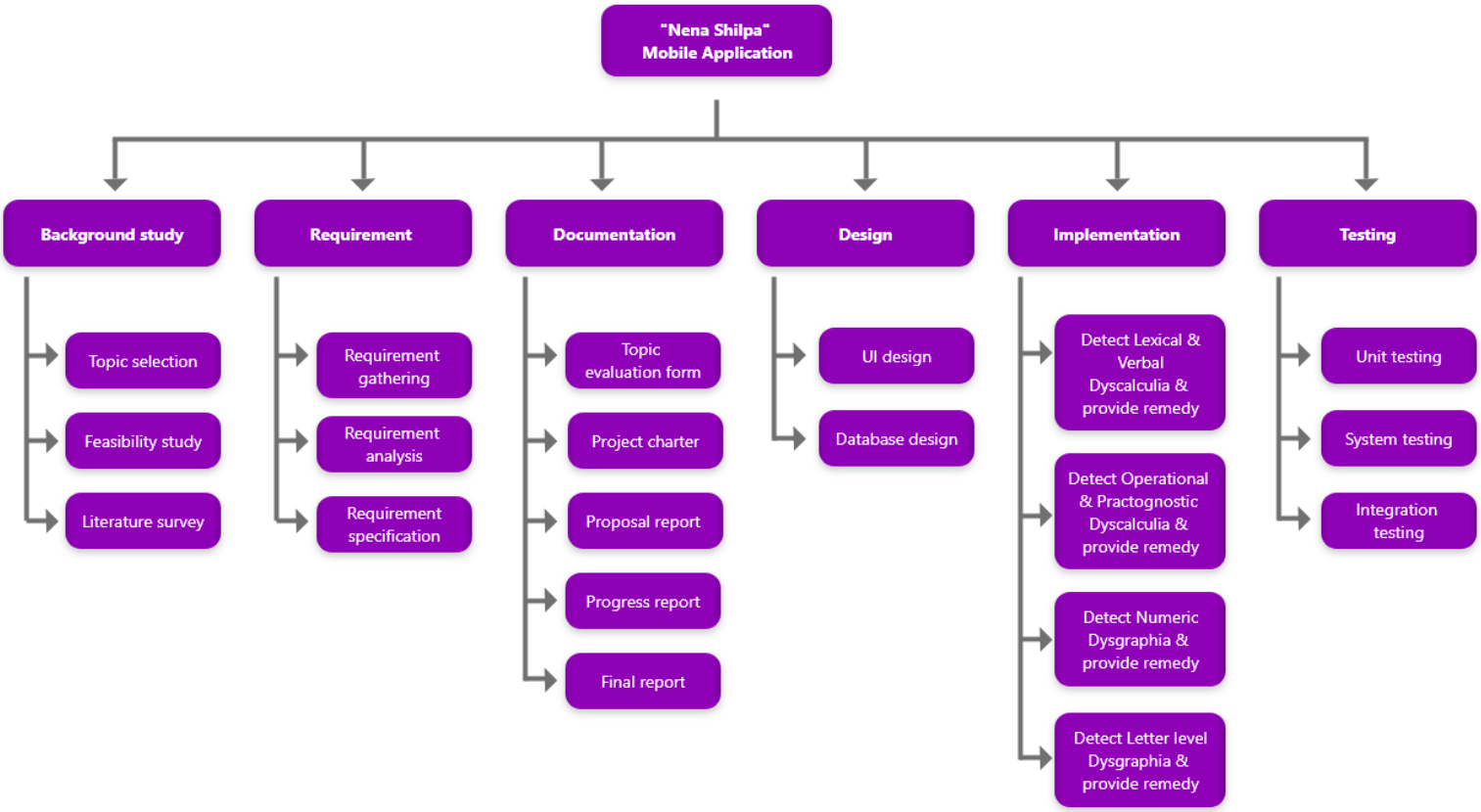
Appendix - B: Flow Diagram



Appendix - C: Gantt Chart



Appendix - D: Work Breakdown Chart



Appendix - E: Recommendation Letter from School Principal



WP/ HO/ILUKOVITA K.V

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තුම්මෝදර දුරකතන අංකය : 036-5682604

To whom it may concern,

I have gone through the mobile application NANA SHILPA: Screening and Refinement System to Identify the Risk of Dyscalculia and Dysgraphia Learning Disabilities in Primary School Students. I can state that the mobile application is very helpful for the Dyscalculia and Dysgraphia Learning Disabilities intervention in Sri Lanka due to its novel approach. I would like to recommend it to more schools for their usage also and wish them to develop the tool in further levels.

Thank you

.....

The principal

Ilukowita K.

Thummodara.

PRINCIPAL
K.WASANTHA UDAYASIR
S.L.P.S III
WP/HO ILUKOWITA K.V
THUMMODARA

Appendix - F: Recommendation Letter from Medical Supervisor

To whom it may concern,

I have personally gone through the mobile application NANA SHILPA: Screening and Refinement System to Identify the Risk of Dyscalculia and Dysgraphia Learning Disabilities in Primary School Students which is implemented by the students Hewapathirana C.D.D. , Kalpani Sandupaba Abeysinghe A. D., Maheshani H.M.P. and Liyanage T.J.P.S.I can recommend this mobile application for children with special educational needs having to learn Dyscalculia and Dysgraphia.

Thank you

Dr. W. M. K. WANIGASINGHE
(M.B.B.S.)
S.L.M.C. Reg. No: 11614
MALABE

.....

Dr. W.M. K. Wanigashinghe

Ministry of Health

Colombo 10

Appendix - G: Doctor Confirmation Letter

To whom it may concern, 19 July 2020

"Nana Shilpa – Screening and refinement system to identify the risk of Dyscalculia and Dysgraphia learning disabilities in students" research project which is conducted by the Hewapathirana C.D.D [IT17182638], A.D Kalpani Sandupaba Abeysinghe [IT17180108], Maheshani H.M.P [IT17182324] and T.J.P.S Liyanage [IT17186766] 4th year students of Sri Lanka Institute of Information Technology under supervision of Prof. Samantha Tellijjagoda.

This is to certify that for the above-mentioned research project, I will be providing the medical consultation for all the medical related aspects in this project. I hereby confirm that as an external supervisor of the project, I will be offering my consultation as a medical officer throughout this project.

Thank you.

Dr. Kamalani Wanigasinghe

Dr. W. M. K. WANIGASINGHE
(M.B.B.S.)
S.L.M.C. Reg. No: 11614
MALABE

Appendix - H: Confirmation for data collection of the school

WP/ HO/ ILUKKOVITA K.V.

ඔජ /හෝ ඉලුක්ඛිවිට කනිෂ්ඨ විද්‍යාලය

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දුරකතන අංකය : 036-5682604

To whom it may concern, 21 July 2020

"Nana Shilpa – Screening and refinement system to identify the risk of Dyscalculia and Dysgraphia learning disabilities in primary school students" research project which is conducted by the Hewapathirana C.D.D [IT17182638], A.D Kalpani Sandupaba Abeysinghe [IT17180108], Maheshani H.M.P [IT17182324] and T.J.P.S Liyanage [IT17186766] 4th year students of Sri Lanka Institute of Information Technology under supervision of Prof. Samantha Tellijjagoda.

This is to certify that for the above-mentioned research project, as the principal of the WP/HO Illukowita Kanishta Vidyalaya, Thumodara I have given my approval for the data collection related to the project from the students and teachers in the school.

Thank you.

Mr. K. Wasantha Udayasiri,

The Principle,

Illukowita Kanishta Vidyalaya,

Thumodara.

PRINCIPAL
K.K.WASANTHA UDAYASIRI
S.L.P.S III
WP/HO ILUKOWITA K.V
THUMMODARA

Appendix - I: Questionnaire for primary school teachers



ඉගෙනුම් ආබාධ (Learning Disabilities) සහිත සිසුන් හඳුනා ගැනීම සඳහා හඳුන්වා දෙන "නැණ ශිල්ප" ජංගම දුරකතන යෙදවුම් පරේෂණ ව්‍යාපෘතියේ තොරතුරු රැස් කිරීම..

Dyscalculia (ඩිස්කැල්කියුලියා) ආබාධය සහිත සිසුන්ගේ ප්‍රධාන ගැටලුව වනුයේ ගණිත කර්ම හා ඉලක්කම් නිවැරදිව හඳුනා ගැනීමට නොහැකි වීමයි.

Dysgraphia (ඩිස්ග්‍රාෆියා) ඉගෙනුම් ආබාධය සහිත සිසුන්ගේ ප්‍රධාන ගැටලුව වනුයේ සපයන ලද ඉඩ ප්‍රමාණය තුල නිවැරදිව අකුරු ලිවීමට නොහැකි වීමයි.

* Required

ඉහත දැක්වෙන ඉගෙනුම් ආබාධ පිලිබඳව ඔබ මීට පෙර දැන සිටියේද? *

- ඔව්
- නැත

ඉහත දැක්වෙන ඉගෙනුම් ආබාධ වල දැක්වෙන ප්‍රධාන ගැටලු ඔබ මීට පෙර අත්දැකීමට ලක්වී හෝ සමීපතමයෙකු තුලින් දැක තිබේද? *

- අත්දැකීමට ලක්ව තිබේ
- අත්දැකීමට ලක්ව නැත
- සමීපතමයෙකු තුලින් දැක තිබේ

ඉහත ඉගෙනුම් ආබාධ පූර්ව අවධියේම හඳුනා ගැනීම වැදගත් යැයි ඔබ සිතන්නේද? *

	1	2	3	4	5	
නැත	○	○	○	○	○	ඔව්

ඉහත ඉගෙනුම් ආබාධ පූර්ව අවධියේම හඳුනා ගැනීම සඳහා ජංගම දුරකතන යෙදවුමක් සුදුසු යැයි ඔබ සිතන්නේද? *

	1	2	3	4	5	
සුදුසු නැත	○	○	○	○	○	සුදුසුයි

ඉහත ඉගෙනුම් ආබාධ පිලිබඳ සමාජය අවශය නරම් දැනුවත් හෝ අවධානය ලබා දී ඇතැයි ඔබ සිතන්නේද? *

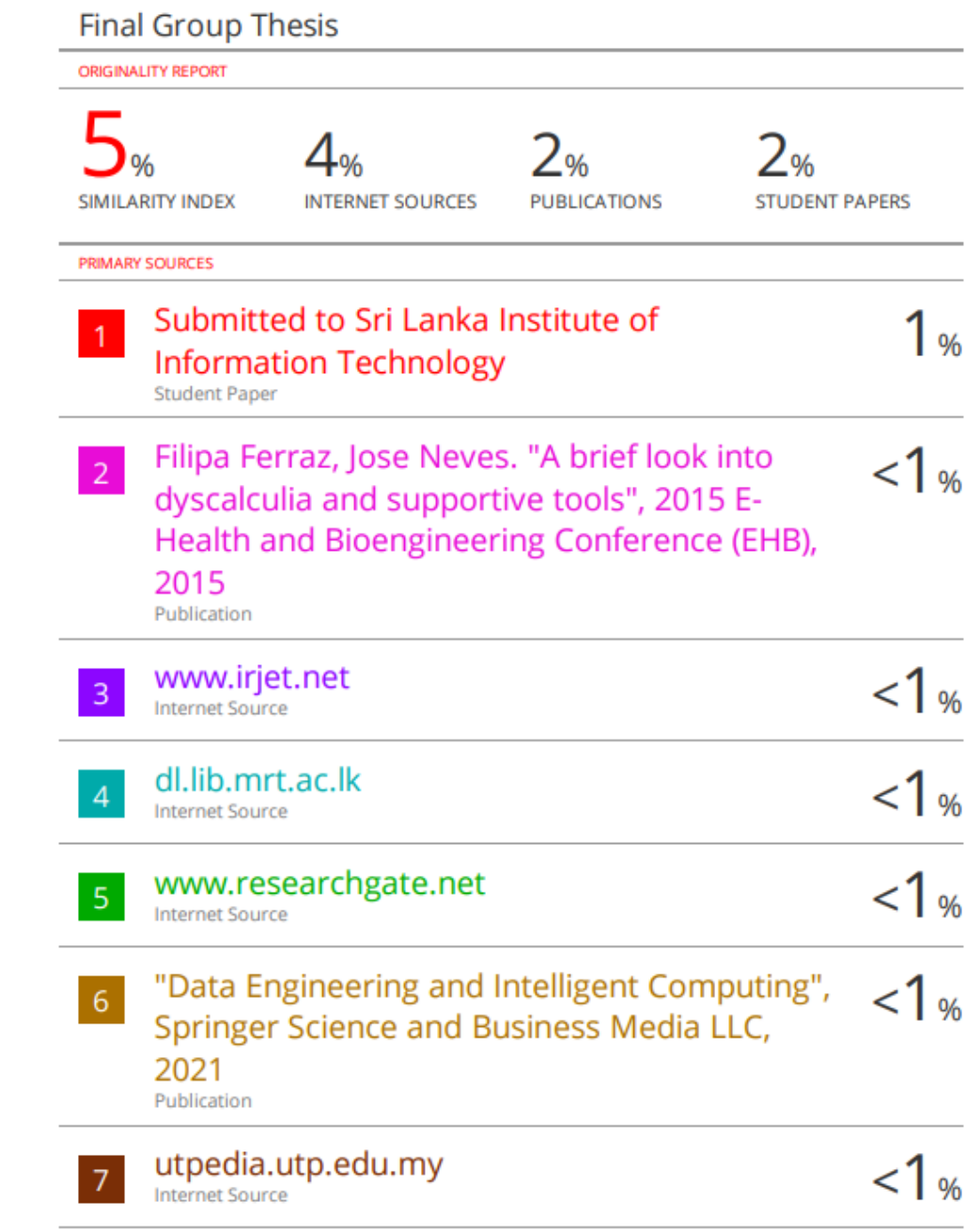
	1	2	3	4	5	
දැනුවත් නැත	○	○	○	○	○	දැනුවත්ව ඇත

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Appendix - J: Plagiarism Report



Final Group Thesis

ORIGINALITY REPORT

5%	4%	2%	2%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Sri Lanka Institute of Information Technology Student Paper	1%
2	Filipa Ferraz, Jose Neves. "A brief look into dyscalculia and supportive tools", 2015 E-Health and Bioengineering Conference (EHB), 2015 Publication	<1%
3	www.irjet.net Internet Source	<1%
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